Bicuspid aortic stenosis: National three-year outcomes of transcatheter versus surgical aortic valve replacement among Medicare beneficiaries

Qiudong Chen, MD,[a] Jad Malas, MD,[a] Dominick Megna, MD,[a] Derrick Y. Tam, MD,[a] George Gill, MD,[a] Georgina Rowe, MD,[a] Sharmini Premananthan, BS,[a] Aasha Krishnan, BS,[a] Achille Peiris, MSHS,[a] Dominic Emerson, MD,[a] Aakriti Gupta, MD,[b] Pedro Catarino, MD,[a] Natalia Egorova, PhD,[c] Joanna Chikwe, MD,[a] and Michael E. Bowdish, MD, MS[a]

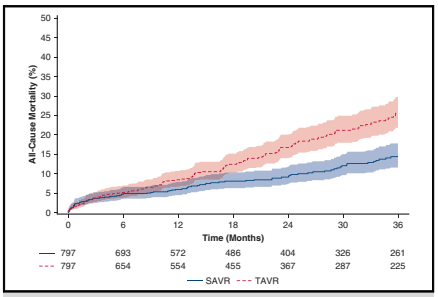
ABSTRACT

Objective: Randomized trials of transcatheter versus surgical aortic valve replacements have excluded bicuspid anatomy. We compared 3-year outcomes of transcatheter aortic valve replacement versus surgical aortic valve replacement in patients aged more than 65 years with bicuspid aortic stenosis.

Methods: The Centers for Medicare and Medicaid data were used to identify 6450 patients undergoing isolated surgical aortic valve replacement (n = 3771) or transcatheter aortic valve replacement (n = 2679) for bicuspid aortic stenosis (2012-2019). Propensity score matching with 21 baseline characteristics including frailty created 797 pairs.

Results: Unmatched patients undergoing transcatheter aortic valve replacement were older than patients undergoing surgical aortic valve replacement (78 vs 70 years), with more comorbidities and frailty (all $P < .001$). After matching, transcatheter aortic valve replacement was associated with a similar mortality risk compared with surgical aortic valve replacement within the first 6 months (hazard ratio [HR], 1.08, 95% CI, 0.67-1.69) but a higher mortality risk between 6 months and 3 years (HR, 2.16, 95% CI, 1.22-3.83). Additionally, transcatheter aortic valve replacement was associated with a lower risk of heart failure readmissions before 6 months (HR, 0.51, 95% CI, 0.31-0.87) but a higher risk between 6 months and 3 years (HR, 4.78, 95% CI, 2.21-10.36). The 3-year risks of aortic valve reintervention (HR, 1.03, 95% CI, 0.30-3.56) and stroke (HR, 1.21, 95% CI, 0.75-1.96) were similar.

Conclusions: Among matched Medicare beneficiaries undergoing transcatheter aortic valve replacement or surgical aortic valve replacement for bicuspid aortic stenosis, 3-year mortality was higher after transcatheter aortic valve replacement. However, transcatheter aortic valve replacement was associated with a similar risk of mortality and a lower risk of heart failure readmissions during the first 6 months after the intervention. Randomized comparative data are needed to best inform treatment choice. (J Thorac Cardiovasc Surg 2024;168:1035-44)



Survival after TAVR versus SAVR for matched patients with bicuspid AS.

CENTRAL MESSAGE

TAVR may be safe in selected patients with bicuspid AS and otherwise favorable anatomy. Randomized trials comparing TAVR to SAVR in patients with bicuspid anatomy are needed to best inform treatment choice.

PERSPECTIVE

Randomized trials of TAVR versus SAVR have excluded bicuspid anatomy. This analysis showed that 3-year mortality was higher after TAVR than matched Medicare beneficiaries undergoing SAVR for bicuspid AS. However, during the first 6 months after the intervention, TAVR was associated with a similar mortality risk and a lower risk of heart failure readmissions. Randomized comparative data are needed.

From the Departments of [a]Cardiac Surgery and [b]Cardiology, Smidt Heart Institute, Cedars-Sinai Medical Center, Los Angeles, Calif; and [c]Department of Population Health Science and Policy, Icahn School of Medicine at Mount Sinai, New York, NY.

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Address for reprints: Joanna Chikwe, MD, Department of Cardiac Surgery, Smidt Heart Institute, Cedars-Sinai Medical Center, 127 S San Vicente Blvd, Pavilion, Suite A3100, Los Angeles, CA 90048 (E-mail: Joanna.Chikwe@cshs.org).

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Abbreviations and Acronyms

AS = aortic stenosis
HR = hazard ratio
ICD-9 = International Classification of Diseases, 9th Revision
ICD-10 = International Classification of Diseases, 10th Revision
IQR = interquartile range
SAVR = surgical aortic valve replacement
TAVR = transcatheter aortic valve replacement

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An estimated 15 to 29% of patients undergoing surgical aortic valve replacement (SAVR) for aortic stenosis (AS) have bicuspid aortic valves, and this prevalence is even higher in younger patients.[1-3] Pivotal randomized trials comparing surgical and transcatheter aortic valve replacement (TAVR) have consistently excluded patients with bicuspid anatomy, primarily due to valve morphology associated with the suboptimal expansion of TAVR devices and uncertain efficacy.[4-7] Registry analyses comparing TAVR for patients with bicuspid AS versus tricuspid AS have demonstrated generally equivalent outcomes with newer generations of TAVR valves.[8-12] However, few population-level studies have compared TAVR directly with SAVR in patients with bicuspid AS,[13,14] and reports on 3-year outcomes beyond hospital discharge are particularly lacking. Despite this, TAVR is increasingly used in patients with bicuspid AS and previously accounted for approximately 20% of the off-label TAVR use in the United States.[15] The aim of this study was to evaluate 3-year outcomes of TAVR and SAVR for bicuspid AS in a national cohort.

MATERIAL AND METHODS

Data Source

We conducted a retrospective cohort study comparing 3-year outcomes after SAVR or TAVR in patients with bicuspid AS, using the Centers for Medicaid and Medicare Services administrative data. Patients undergoing isolated SAVR or TAVR in the United States between 2012 and 2019 with AS and bicuspid aortic valve were identified using the International Classification of Diseases, 9th Revision (ICD-9) or International Classification of Diseases, 10th Revision, Clinical Modification (ICD-10) diagnosis and procedure codes (Table E1). Isolated SAVR or TAVR was defined as the lack of other major concomitant cardiovascular procedures such as coronary artery bypass grafting, percutaneous coronary interventions, surgical or transcatheter mitral/tricuspid/pulmonic valve interventions, heart transplantation/durable ventricular assist device implantation, or surgery on the thoracic aorta. The ICD-9 or ICD-10 diagnosis codes documented in the index hospitalization record and all inpatient admissions within the prior 2 years were used to identify baseline comorbidities (Table E2). After further exclusion of patients with infective endocarditis, previous aortic valve interventions, presence of prosthetic heart valves, prior heart transplants or durable ventricular assist devices, and patients without Medicare continuous enrollment or at least 1 year of Medicare coverage, we identified 6450 patients undergoing isolated SAVR (n = 3771) or TAVR (n = 2679) (Figure 1). The Centers for Medicaid and Medicare Service Data Protection Review Board and the Institutional Review Board at Cedars-Sinai Medical Center approved this study with a waiver of informed consent (STUDY00001188, approved on February 19, 2021).

Study End Points

The primary end point was all-cause mortality during a median follow-up time of 2.4 (interquartile range [IQR], 1.0-4.3) years. Secondary outcomes included 30-day rates of mortality, stroke, permanent pacemaker placement, length of index hospitalization, 3-year rates of stroke, aortic valve reintervention, and heart failure readmission. Deaths were identified from the Master Beneficiary Summary File, and non-fatal secondary end points were defined using ICD-9 or ICD-10 diagnosis and procedure codes (Table E3). Stroke was defined as any cerebrovascular accidents documented during the index hospitalization (that was not present on admission) and any subsequent hospital admission where the principal diagnosis was hemorrhagic or ischemic stroke (not including transient ischemic attacks). Aortic valve reintervention included any subsequent transcatheter or surgical procedures involving the aortic valve. Heart failure readmissions included any subsequent hospital admission where the principal diagnosis was heart failure or cardiomyopathy. We censored patients in the absence of death, stroke, aortic valve reintervention, or heart failure readmission on December 31, 2019, or 3 years after the index aortic valve procedure, whichever occurred first.

Statistical Analysis

Baseline patient characteristics were reported as mean ± SD or median with IQR for continuous variables depending on overall distribution and proportions for categorical variables. Between-group comparisons were performed using Student *t* test or Wilcoxon rank-sum test for continuous variables depending on the variable distribution. Pearson's chi-square test was performed for categorical variables.

We performed propensity score matching to create balanced treatment groups for comparisons. A logistic regression model was fitted with TAVR as the dependent variable to calculate the propensity score, incorporating individual hospitals as a fixed effect. Covariates included in the model were age, sex, hypertension, hyperlipidemia, peripheral vascular disease, prior strokes, coronary artery disease, congestive heart failure/cardiomyopathy, chronic kidney disease, dialysis dependency, chronic obstructive pulmonary disease, liver disease, cancer, coagulopathy, immobility, cognitive dysfunction, supplemental oxygen dependency, admission type, procedure year, Charlson Comorbidity Index, and frailty score. The frailty score was generated using a previously published method that allows frailty assessment based on administrative claims data without physical performance measures.[16] The area under the receiver operating curve for this model was 0.91. Next, a 1:1 greedy nearest neighbor match was performed using a caliper of 0.01 of the logit of the propensity score computed by this model. Standardized differences (instead of *P* values) were reported after matching to assess covariate balance. A standardized difference of 10% or less was deemed to be the ideal balance. To assess the risk of residual confounding after propensity score matching, we tested the association between treatment groups

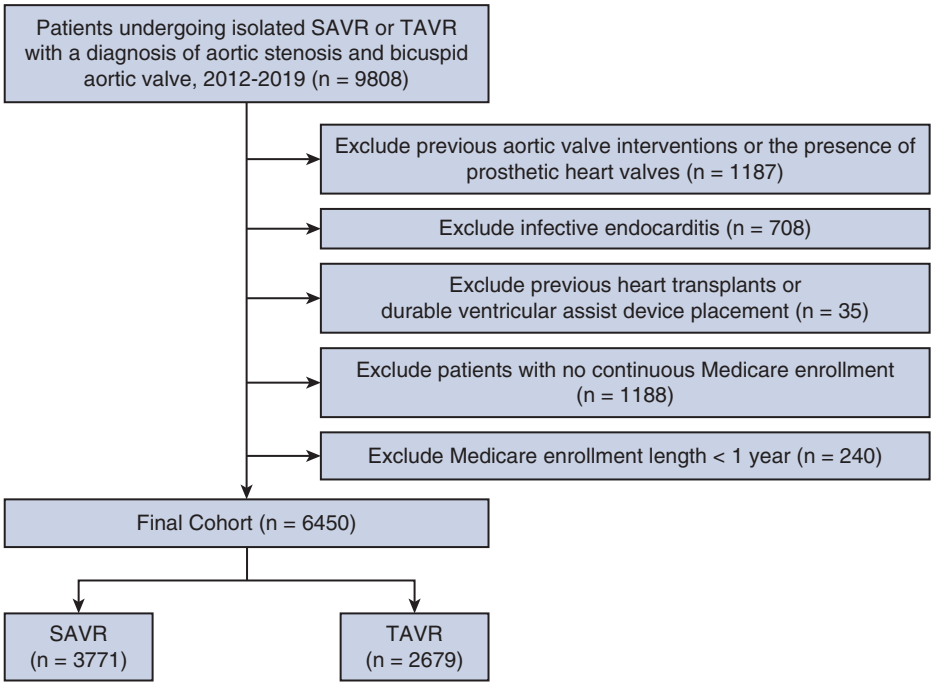


FIGURE 1. Study cohort identification. *SAVR*, Surgical aortic valve replacement; *TAVR*, transcatheter aortic valve replacement.

(SAVR vs TAVR) and a composite falsification end point of hospitalization for urinary tract infection or pneumonia in the matched cohort, defined using ICD-9 or ICD-10 diagnosis codes (Table E3). The cumulative incidence of this composite falsification end point was evaluated with death as a competing risk.

In propensity-matched patients, survival curves were derived from the life table. Thirty-day outcomes were compared using McNemar's test. For the nonfatal secondary end points of stroke, aortic valve reintervention, and heart failure readmission at 3 years, death was considered as a competing risk to construct cumulative incidence function curves and to calculate 3-year estimates. The difference in overall survival was compared using the marginal Cox proportional hazard model with robust sandwich variance estimators, whereas differences in non-fatal secondary end points were evaluated using Fine-Gray sub-distributional hazard models. These models were fitted with only the intervention type (TAVR vs SAVR) entered as a covariate. Violation of the proportional hazard assumption was observed in the analysis of all-cause mortality and heart failure readmissions. This was addressed by including an interaction term between the intervention type and follow-up time, with follow-up time defined as dichotomous. Follow-up time was dichotomized at 6 months, which was chosen on the basis of the shape of the survival or cumulative incidence function curves in matched patients. Landmark analyses were also performed at these cutoff time points.

To ensure the robustness of our results, we performed 2 sensitivity analyses of the primary end point of all-cause mortality. In the first sensitivity analysis, the association between intervention type and all-cause mortality was evaluated using a multivariable Cox model with a robust sandwich variance estimator to account for center clustering using the entire cohort of 6450 patients. Variables included in the model for adjustment were the same as those used to create propensity scores. Violation of the proportional hazard assumption was similarly addressed by including an interaction term between the intervention type and time as mentioned in the preceding text. In the second sensitivity analysis, we used overlap propensity score weighting (on the entire cohort of 6450 patients). This method mimics the attributes of a randomized clinical trial by achieving the exact balance on the mean of every measured covariate, and it may be preferred over conventional inverse probability of treatment weighting when the initial differences in treatment groups are large such as in this study[17] (Appendix E1 and Figure E1).

All tests were 2-tailed with an alpha level of 0.05. No formal adjustment for multiple hypothesis testing was made, and analyses should be interpreted as exploratory. All statistical analyses were performed using SAS 9.4 (SAS Institute, Inc).

RESULTS

Patient Population

After applying the exclusion criteria, 6450 patients with bicuspid AS undergoing isolated SAVR (n = 3771) or TAVR (n = 2679) were identified (Figure 2). Five patients received SAVR on the same date as or after TAVR during the same index hospitalization, and they were included in the TAVR group for analysis. At baseline, patients who underwent TAVR were older (78 [IQR, 71-84] vs 70 [67-74] years, $P < .001$), had a higher frailty score (0.14 [IQR, 0.08-0.24] vs 0.06 [IQR, 0.04-0.09], $P < .001$), and were more likely to have significant comorbidities (Table 1). After propensity score matching, 797 patient pairs were identified. Baseline characteristics stratified by intervention type before and after propensity matching are shown in Table 1. Propensity matching yielded groups with similar baseline characteristics (all standardized mean differences <10%). Association between procedure year and the choice of SAVR versus TAVR procedures before and after matching are shown in Table E4. The 3-year cumulative incidence of the falsification end point (hospitalization for urinary tract infection or

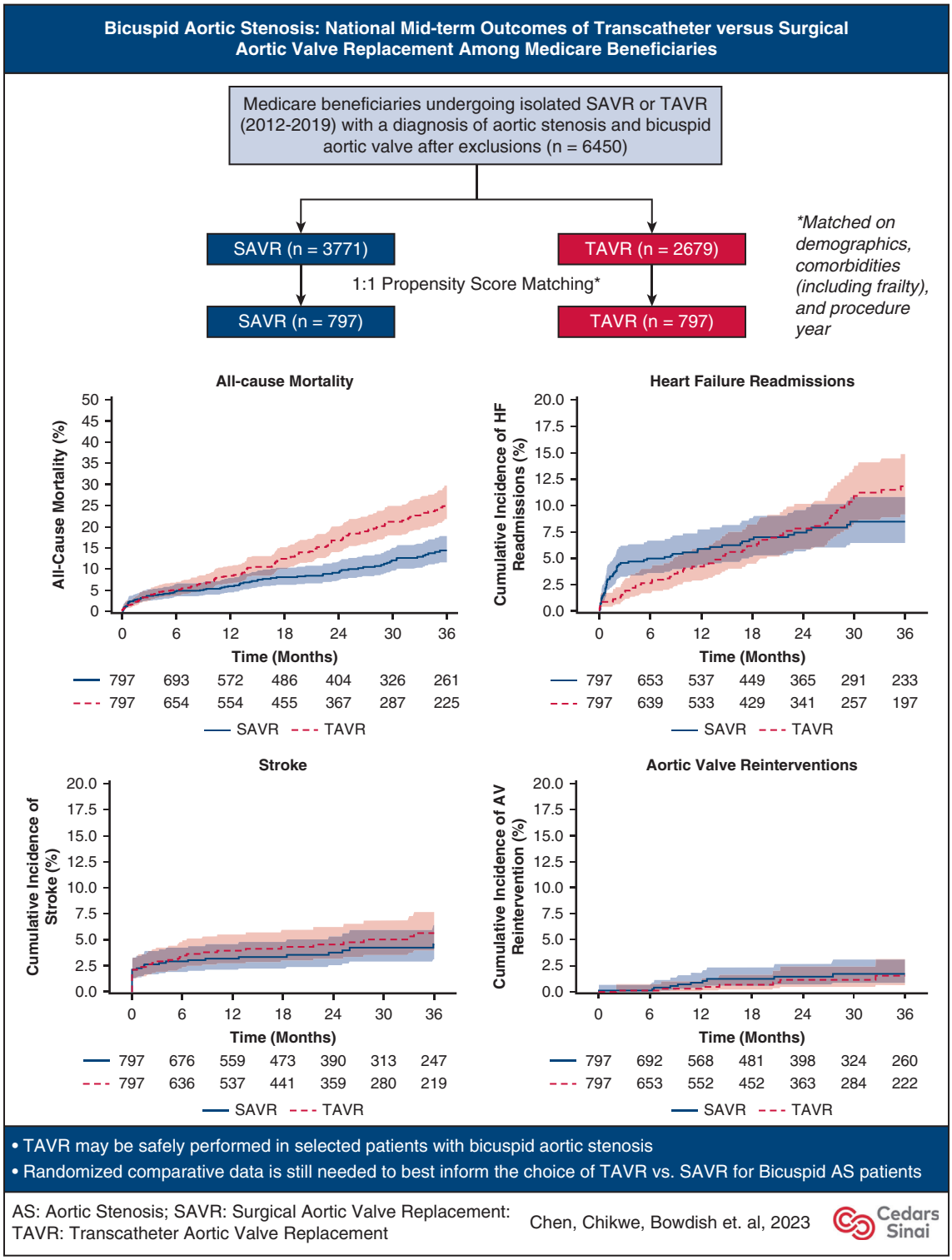


FIGURE 2. Three-year outcomes of TAVR versus SAVR among Medicare beneficiaries with bicuspid AS.

TABLE 1. Baseline characteristics in patients with bicuspid aortic stenosis undergoing isolated surgical or transcatheter aortic valve replacement

Variable	Unmatched			1:1 Propensity matched		
	SAVR (n = 3771)	TAVR (n = 2679)	*P* value	SAVR (n = 797)	TAVR (n = 797)	SMD (%)
Age	70 (67-74)	78 (71-84)	<.001	73 (69-77)	73 (68-80)	3.9
Male sex	60.2 (2270)	56.9 (1523)	.007	58.2 (464)	57.3 (457)	−1.8
White race	91.9 (3466)	92.4 (2475)	.06	92.6 (738)	93.6 (746)	3.7
Hypertension	87.9 (3313)	95.4 (2555)	<.001	91.6 (730)	91.6 (730)	0.0
Hyperlipidemia	81.0 (3055)	89.4 (2396)	<.001	84.4 (673)	83.8 (668)	−1.8
Diabetes	31.6 (1192)	46.9 (1255)	<.001	37.1 (295)	40.9 (326)	8.1
Peripheral vascular disease	27.6 (1040)	53.6 (1435)	<.001	40.7 (324)	40.8 (325)	0.3
Cerebrovascular disease	30.3 (1141)	45.8 (1228)	<.001	37.0 (295)	37.8 (301)	1.6
Prior stroke	4.7 (176)	10.0 (267)	<.001	7.3 (58)	7.7 (61)	1.4
Atrial fibrillation	25.8 (972)	42.3 (1134)	<.001	32.5 (259)	33.3 (265)	1.6
Coronary artery disease	67.8 (2555)	87.3 (2339)	<.001	80.0 (635)	79.9 (637)	0.6
Myocardial infarction	14.2 (534)	32.4 (868)	<.001	21.9 (175)	20.6 (164)	−3.3
Previous CABG	4.3 (163)	18.3 (491)	<.001	8.4 (67)	9.3 (74)	2.8
Previous PCI	12.1 (457)	34.5 (923)	<.001	20.8 (166)	20.8 (166)	0.0
Congestive heart failure	54.4 (2053)	85.2 (2283)	<.001	73.9 (589)	72.7 (579)	−2.9
Chronic obstructive pulmonary disease	26.6 (1005)	44.7 (1197)	<.001	35.1 (280)	33.3 (265)	−4.0
Chronic kidney disease	21.4 (807)	46.1 (1236)	<.001	33.0 (263)	32.4 (258)	−1.3
Dialysis dependency	2.0 (77)	5.5 (147)	<.001	3.0 (24)	3.0 (24)	0.0
Liver disease	16.3 (615)	27.0 (724)	<.001	22.1 (176)	21.6 (172)	−1.2
Coagulation disorder	15.5 (584)	21.7 (580)	<.001	18.3 (146)	18.2 (145)	−0.3
Cancer	25.3 (955)	40.7 (1089)	<.001	31.9 (254)	33.3 (265)	3.0
Immobility	9.6 (362)	26.7 (714)	<.001	16.1 (128)	14.4 (115)	−4.3
Cognitive dysfunction	5.7 (213)	15.8 (422)	<.001	9.4 (75)	9.2 (73)	−0.8
Supplemental oxygen dependency	1.8 (68)	11.2 (299)	<.001	4.8 (38)	4.6 (37)	−0.5
Elective admission	90.8 (3425)	85.4 (2289)	<.001	89.3 (712)	89.1 (710)	1.2
Procedural year	2016 (2014-2018)	2017 (2016-2019)	<.001	2017 (2015-2018)	2017 (2015-2018)	1.6
Charlson Comorbidity Index	2 (1-3)	4 (2-6)	<.001	3 (1-5)	3 (1-5)	−0.8
Frailty score	0.06 (0.04-0.09)	0.14 (0.08-0.24)	<.001	0.08 (0.05-0.14)	0.09 (0.06-0.15)	1.7

Variables are expressed in % (n) or median (IQR). *SAVR*, Surgical aortic valve replacement; *TAVR*, transcatheter aortic valve replacement; *SMD*, standardized mean difference; *CABG*, coronary artery bypass grafting; *PCI*, percutaneous coronary intervention.

pneumonia) was similar for SAVR (6.8%, 95% CI, 4.7-9.3) and TAVR (7.1%, 95% CI, 5.0-9.7, $P = .80$, Figure E2).

Primary End Point

The comparison of all-cause mortality in the unmatched cohort is shown in Figure E3. Thirty-day mortality in the unmatched cohort was 1.9% after SAVR and 2.5% after TAVR ($P = .10$). In the matched population, all-cause mortality at 6, 12, 24, and 36 months was 4.8% (95% CI, 3.5-6.6), 6.0% (95% CI, 4.5-8.0), 9.2% (95% CI, 7.2-11.7), 14.5% (95% CI, 11.7-17.9) after SAVR and 5.2% (95% CI, 3.9-7.1), 8.5% (95% CI, 6.6-10.8), 16.7% (95% CI, 14.0-20.0), 25.4% (95% CI, 21.7-29.6) after TAVR, respectively (Figure 3). The proportional hazard assumption was violated due to a delayed effect, as the all-cause mortality curves for TAVR and SAVR patients did not begin to separate until after 6 months. TAVR was associated with a similar mortality risk compared with SAVR within the first 6 months (hazard ratio [HR], 1.08, 95% CI, 0.67-1.69, $P = .74$) but a higher mortality risk after 6 months (HR, 2.16, 95% CI, 1.22-3.83, $P = .008$). Landmark analysis showed similar findings (Figure 4, *A*). In the sensitivity analyses of the primary end point using multivariable Cox modeling (Table E5), TAVR was also associated with a similar mortality risk compared with SAVR within the first 6 months (HR, 1.24, 95% CI, 0.95-1.60, $P = .10$) but a higher mortality risk after 6 months (HR, 2.50, 95% CI, 1.90-3.28, $P < .001$). A second sensitivity analysis using overlap propensity score weighting again

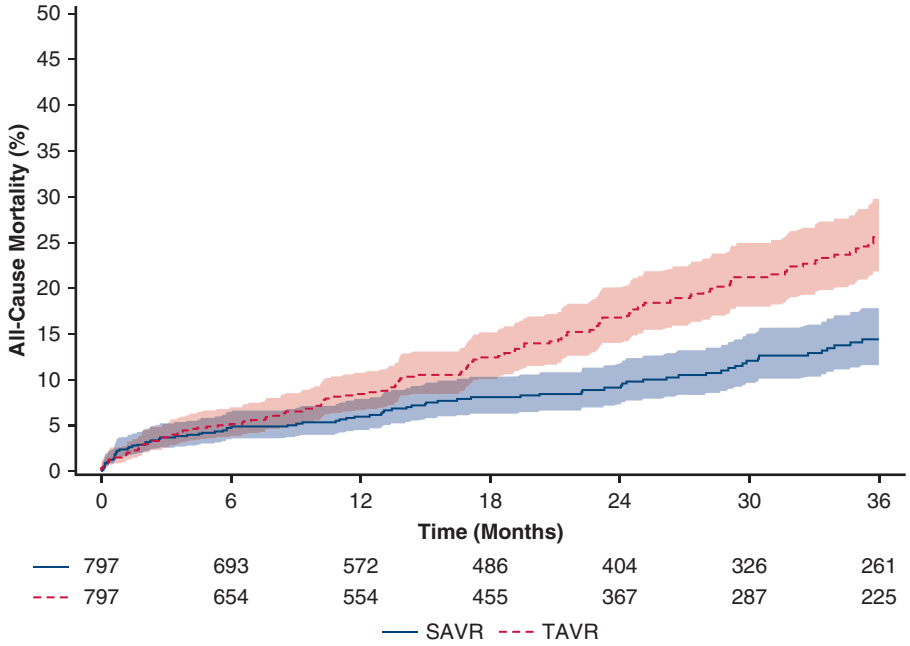


FIGURE 3. All-cause mortality after isolated SAVR versus TAVR for bicuspid AS in propensity-matched patients. 95% CIs are shown. *SAVR*, Surgical aortic valve replacement; *TAVR*, transcatheter aortic valve replacement.

produced similar findings: TAVR was also associated with a similar mortality risk compared with SAVR within the first 6 months (HR, 1.04, 95% CI, 0.77-1.41, $P = .80$) but a higher mortality risk after 6 months (HR, 2.42, 95% CI, 1.62-3.61, $P < .001$).

Secondary End Points

Unmatched 30-day stroke rates were 1.9% after SAVR and 2.1% after TAVR ($P = .59$). Unmatched 30-day new permanent pacemaker implantation rate was 5.9% after SAVR and 9.3% after TAVR ($P < .001$). After matching, 30-day mortality was 2.4% after SAVR and 1.6% after TAVR ($P = .29$). Thirty-day stroke rate was 2.3% after SAVR and 2.4% after TAVR ($P = .87$). Thirty-day new permanent pacemaker implantation rate was 6.3% after SAVR and 9.2% after TAVR ($P = .03$), a difference mostly driven by the higher permanent pacemaker implantation rate with TAVR during the earlier study period (Table E6). Patients undergoing TAVR also had a shorter index hospitalization than SAVR patients (3 [IQR, 2-6] vs 7 [IQR, 5-10] days, $P < .001$).

Unmatched comparisons of secondary outcomes are presented in Figure E4. In matched patients, the 3-year cumulative incidence of heart failure readmissions was 8.5% (95% CI, 6.5-10.8) after SAVR and 11.8% (9.2-14.9) after TAVR (Gray's test $P = .51$, Figure 5, *A*). Nonproportionality was observed during follow-up as the cumulative incidence curves crossed: TAVR was associated with a lower risk of heart failure readmissions than SAVR during the first 6 months after the index procedure (HR, 0.51, 95% CI, 0.31-0.87, $P = .01$) but a higher risk afterward (HR, 4.78, 95% CI, 2.21-10.36, $P < .001$). Landmark analysis showed similar findings (Figure 4, *B*). Three-year aortic valve reintervention rates were similar for SAVR (1.7%, 95% CI, 0.9-3.1) and TAVR (1.6%, 95% CI, 0.7-3.2) (HR, 0.75, 95% CI, 0.28-1.97, $P = .55$, Figure 5, *B*). Stroke rates at 3 years were similar after SAVR (4.6%, 95% CI, 3.1-6.4) and TAVR (5.6%, 95% CI, 4.0-7.7) (HR, 1.21, 95% CI, 0.75-1.96, $P = .42$, Figure 5, *C*).

DISCUSSION

This analysis of Medicare beneficiaries with a median age of 73 years undergoing isolated SAVR and TAVR for bicuspid AS has 3 key findings. First, TAVR may be safely performed in selected patients with bicuspid AS: In propensity-matched patients, the observed 30-day mortality was 1.6% compared with 2.4% after SAVR. Second, SAVR was associated with greater 3-year survival and freedom from hospital readmissions compared with TAVR in matched patients. Third, stroke and aortic valve reintervention risks associated with TAVR at 3 years were similar to that after SAVR.

Direct comparisons of TAVR and SAVR in the bicuspid population are limited to observational studies, and only perioperative outcomes have been reported. Consistent with our findings, previous studies have demonstrated a similar 30-day rate of mortality and stroke after risk adjustment, suggesting the safety of TAVR for selected bicuspid AS in contemporary practice.[13,14,18] However, TAVR was

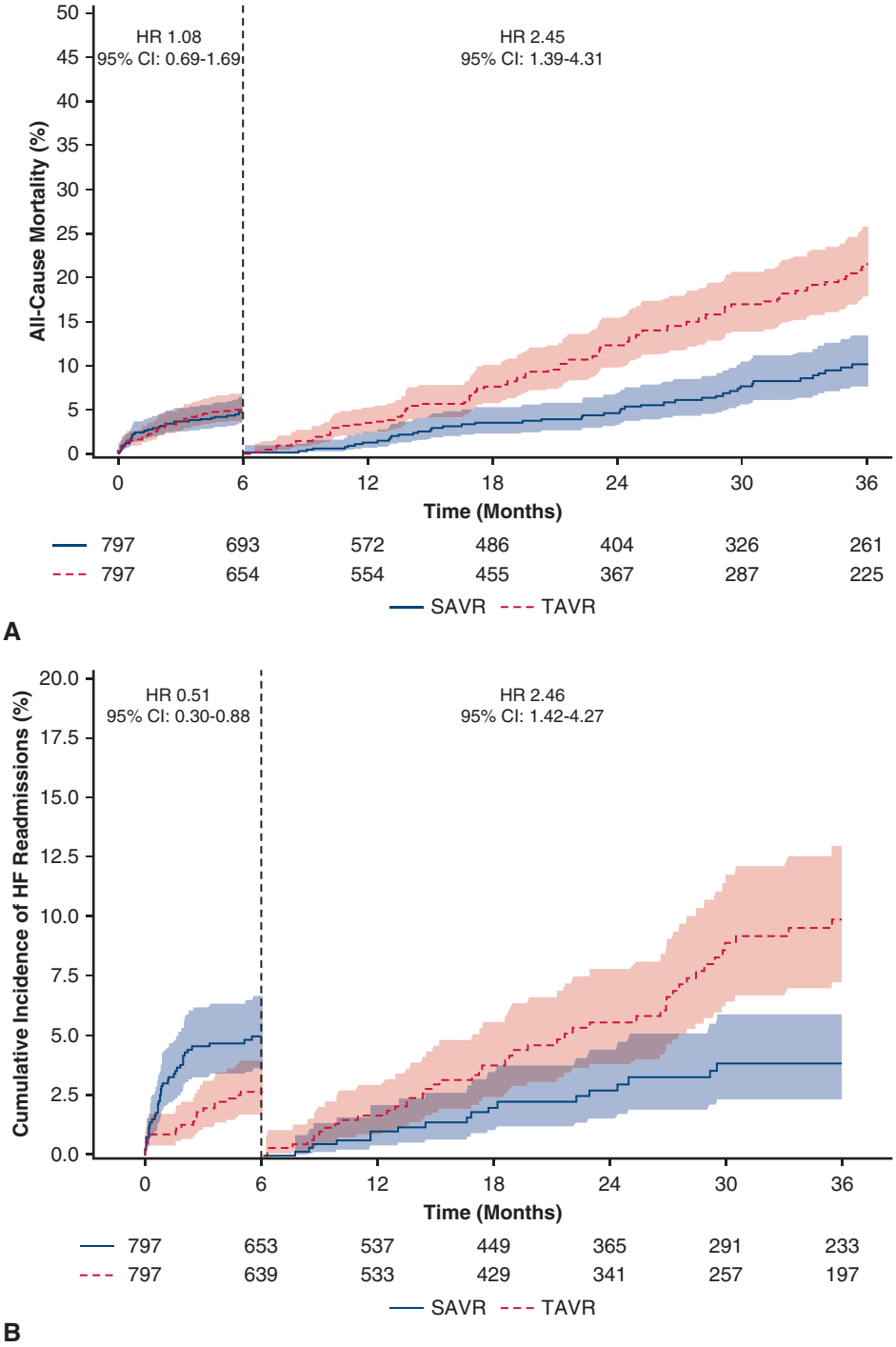


FIGURE 4. Landmark analysis of all-cause mortality (A) and heart failure readmissions (B) after isolated SAVR versus TAVR (with 95% CI) for bicuspid AS in matched patients. *SAVR*, Surgical aortic valve replacement; *TAVR*, transcatheter aortic valve replacement.

consistently associated with a significantly higher rate of new permanent pacemaker placement than SAVR (9.2% vs 6.3% in the present study, $P = .03$), and this risk should be considered especially when evaluating young patients with bicuspid anatomy with a low surgical risk for TAVR versus SAVR.

Multiple observational studies have also demonstrated comparable outcomes of TAVR in patients with bicuspid versus tricuspid AS.[4,8-11] In an analysis of the Society of

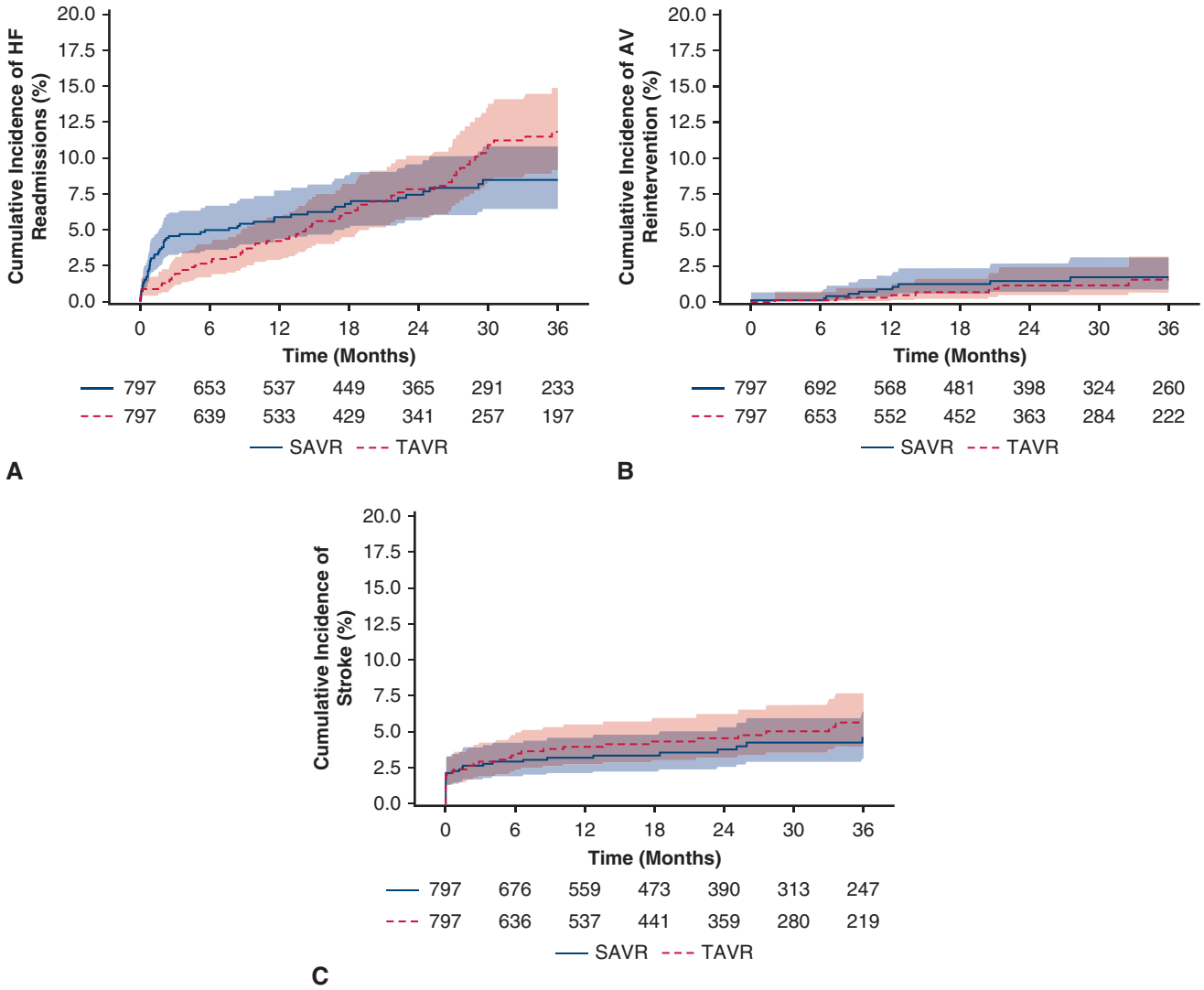


FIGURE 5. Secondary outcomes after isolated SAVR versus TAVR for bicuspid AS in propensity-matched patients. Cumulative incidence (with 95% CIs) of heart failure readmissions (A), aortic valve reintervention (B), and stroke (C) are shown. Death was treated as a competing risk event in these analyses. *SAVR*, Surgical aortic valve replacement; *TAVR*, transcatheter aortic valve replacement.

Thoracic Surgeons/American College of Cardiology Transcatheter Valve Therapy registry, using third-generation balloon-expandable TAVR devices for bicuspid versus tricuspid AS had no significant difference in 30-day (2.6% vs 2.5%, $P = .82$) or 1-year mortality (10.5% vs 12.0%, $P = .31$) after propensity matching. There were also no significant differences in the incidence of moderate or severe paravalvular leak at 30 days. However, a higher 30-day stroke rate was observed in matched patients with bicuspid anatomy (2.5% vs 1.6%, $P = .02$).[10] Similar comparisons of TAVR for bicuspid versus tricuspid AS were performed using data from the Placement of Aortic Transcatheter Valves 3 bicuspid registry, with no observed differences in 1-year composite rate of death, stroke, and cardiovascular rehospitalization (bicuspid 10.9% vs tricuspid 10.2%, $P = .80$).[8] Additionally, in a multicenter, propensity-matched analysis comparing low-risk patients with bicuspid AS to low-risk patients with tricuspid aortic stenosis from the TAVR arm of the Evolut low-risk trial, the rates of the primary composite outcome of all-cause mortality, disabling stroke, or aortic valve–related hospitalization at 1 year were similar (bicuspid 4.1% vs tricuspid 6.9%; $P = .30$).[11] Despite these findings, in the absence of randomized data with a surgical arm for comparison or longer follow-ups, there is insufficient evidence to conclude that TAVR may be similar to SAVR in patients with bicuspid anatomy.

The optimal treatment for patients with bicuspid aortic valves and severe symptomatic AS is debated. Current consensus guidelines recommend that TAVR may be considered as an alternative to SAVR after taking into account patient-specific procedural risks, values, tradeoffs,

and preferences, and when surgery is to be performed at a Comprehensive Valve Center (Class 2b, level of evidence B).[19] Our results provide valuable insights into the 3-year outcomes of TAVR and SAVR in patients with bicuspid AS, which has not been reported previously. That said, the potential for residual confounding should be considered, especially given the lack of information in Medicare data regarding the bicuspid valve phenotype, detailed echocardiographic variables, pattern or degree of associated aortopathy, and procedural details such as the use of cerebral protection devices, approach, implantation depth, and post-implantation dilation. This suggests that despite our efforts with propensity score matching, the SAVR and TAVR groups being compared may still be very different. The significant differences between treatment groups before matching may also suggest that treatment choice seems appropriately tailored according to individual patient risk profile. Our findings also contrasted with randomized trials demonstrating largely equivalent outcomes of SAVR and TAVR in patients with tricuspid AS,[4-7] although patients with bicuspid anatomy were excluded from these trials. Additionally, considering the elderly age and significant comorbidities present, our matched study cohort likely represents patients with at least intermediate to high surgical risk. Based on recent Transcatheter Valve Therapy registry data, the mean Society of Thoracic Surgeons–predicted risk of mortality for all patients with bicuspid anatomy undergoing TAVR was 4.9%.[10] Therefore, caution should be exercised before extrapolating these findings to a younger, low-risk population with bicuspid anatomy, among whom the choice between TAVR and SAVR may be the most relevant.

Limitations

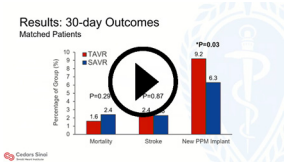
Despite using national data to compare 3-year outcomes of TAVR and SAVR in a relatively contemporary population with bicuspid AS, our study has several limitations. First and most important, there was insufficient information regarding important clinical characteristics such as the severity of comorbidities, presence of clinically significant coronary artery disease, anatomic phenotype of the bicuspid aortic valve, degree, and pattern of calcification of aortic valve, measurement of the aortic root apparatus, coronary take-off, postprocedural echocardiographic outcomes, and the degree of associated aortopathy that may affect longer-term outcomes. This increases the difficulty of controlling for the significant selection bias and may explain some of the discrepancies between the outcomes observed in our analysis. Second, the matched cohort contained only 25% of the overall cohort, suggesting it was difficult to find comparable SAVR and TAVR patients with bicuspid AS. Consequently, the matched population may not accurately represent the risk profile of patients with bicuspid anatomy undergoing SAVR or TAVR in the real world, limiting the external validity of our results. Third, administrative data may have limited accuracy in identifying patients with bicuspid AS,[20,21] and it is possible that severely calcified trileaflet valves appear bicuspid on imaging, leading to possible misclassification of valve morphology in Medicare data. However, we believe that combining the ICD diagnosis codes for AS and bicuspid aortic valve with the ICD procedure codes for SAVR and TAVR likely produces a relatively high degree of accuracy. Multiple previous analyses of administrative datasets have also used a similar methodology.[13,14,22,23] Additionally, because of the possibility that only a subset of patients with bicuspid AS undergoing intervention were identified, we avoided evaluating procedural trends over time. Fourth, we could not identify the type of TAVR devices used. Because the study period spanned 7 years, multiple generations of TAVR devices were likely included. Fifth, because only Medicare beneficiaries were included, our findings are not generalizable to a younger population, among whom the choice between TAVR and SAVR may be the most relevant. Further studies are needed to elucidate comparative outcomes of TAVR and SAVR for patients aged less than 65 years with bicuspid anatomy. Last, data on causes of death or valve integrity were not available.

CONCLUSIONS

This analysis of TAVR and SAVR for bicuspid AS in national practice suggests that TAVR may be performed safely in selected patients with bicuspid AS with otherwise favorable anatomy, with observed 30-day mortality of 1.6% in a subgroup of patients who were potentially eligible for either therapy. In matched patients, SAVR was associated with greater survival and freedom from hospital readmissions, and similar stroke and aortic valve reintervention compared with TAVR at 3 years. These differences in 3-year survival and heart failure readmissions support the need for randomized trials to inform treatment choice.

Webcast

You can watch a Webcast of this AATS meeting presentation by going to: https://www.aats.org/resources/bicuspid-aortic-stenosis-national-3-year-outcomes-of-tavr-versus-savr.



Conflict of Interest Statement

The authors reported no conflicts of interest.

The *Journal* policy requires editors and reviewers to disclose conflicts of interest and to decline handling or reviewing manuscripts for which they may have a conflict of interest. The editors and reviewers of this article have no conflicts of interest.

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Key Words: aortic stenosis, bicuspid aortic valve, surgical aortic valve replacement, transcatheter aortic valve replacement

APPENDIX E1. OVERLAP PROPENSITY SCORE WEIGHTING

We used overlap weighting on the entire cohort of 6450 patients as a method of sensitivity analysis of the primary end point of all-cause mortality. This method assigns weights to each patient that are proportional to the probability of that patient belonging to the opposite treatment group. This results in weights that are smaller for extreme propensity score (PS) values so that outliers who are nearly always treated (PS near 1) or never treated (PS near 0) do not dominate results, as occurs with inverse probability treatment weighting. Consequently, patients whose characteristics are compatible with either treatment contribute relatively more. This method achieves a target population that mimics the characteristics of a pragmatic randomized trial, excluding no participants from the available sample but emphasizing the comparison of patients at clinical equipoise. It also results in exact balance on the mean of every measured covariate. Propensity scores were first calculated similarly to what was described in the main text. Next, TAVR patients were weighted by the probability of receiving SAVR (1 − PS), and SAVR patients were weighted by the probability of receiving TAVR (PS). Subsequently, the primary end point of all-cause mortality was analyzed using a weighted Cox proportional hazards model.

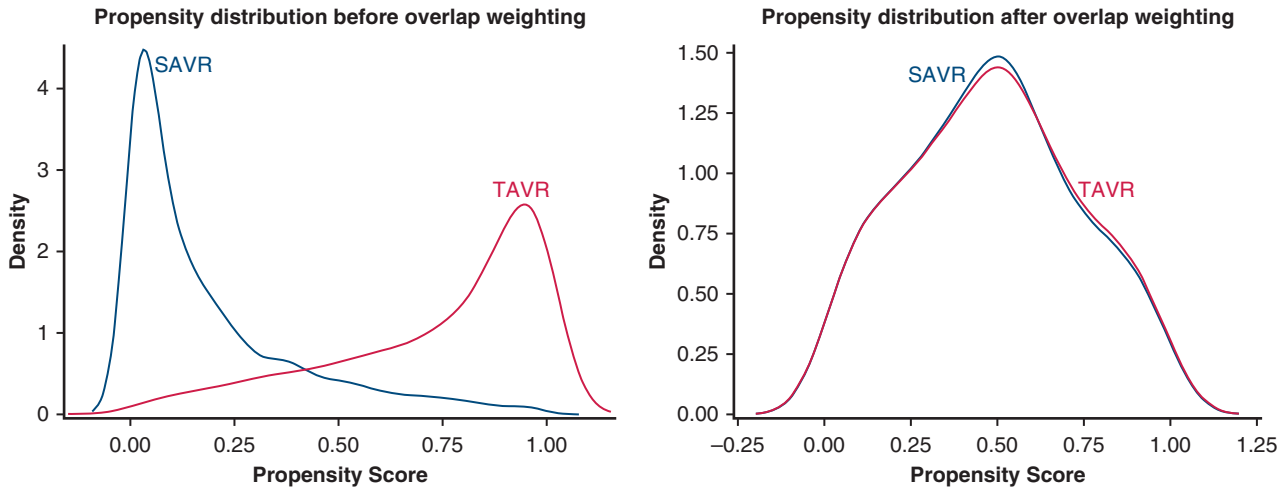

FIGURE E1. Distribution of propensity score before and after overlap weighting. *SAVR*, Surgical aortic valve replacement; *TAVR*, transcatheter aortic valve replacement.

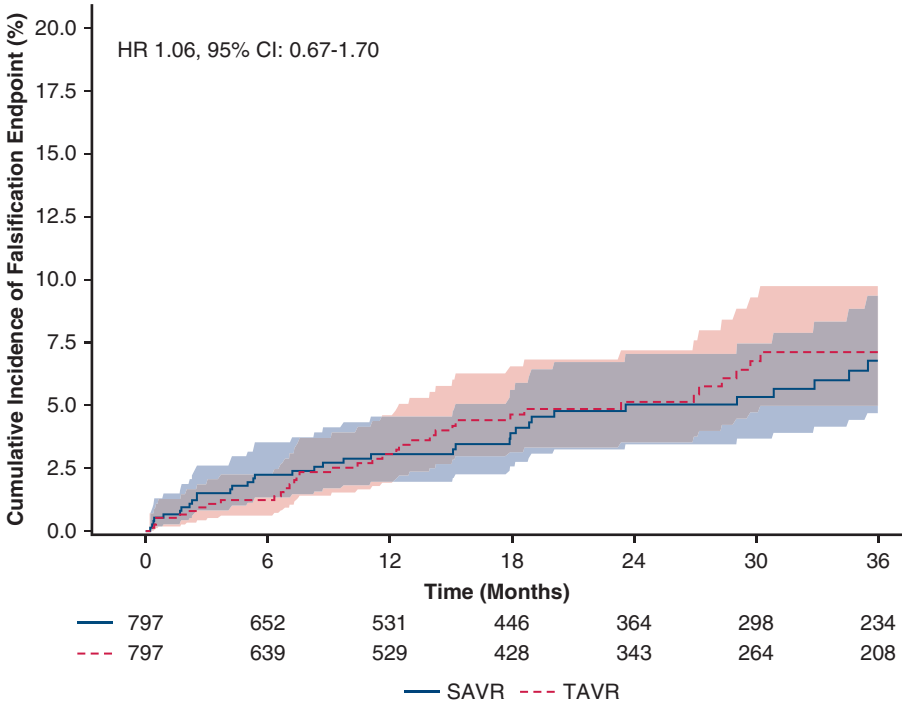

FIGURE E2. Cumulative incidence of the falsification end point of hospitalizations for urinary tract infection or pneumonia in the matched cohort (with 95% CI). *HR*, Hazard ratio; *SAVR*, surgical aortic valve replacement; *TAVR*, transcatheter aortic valve replacement.

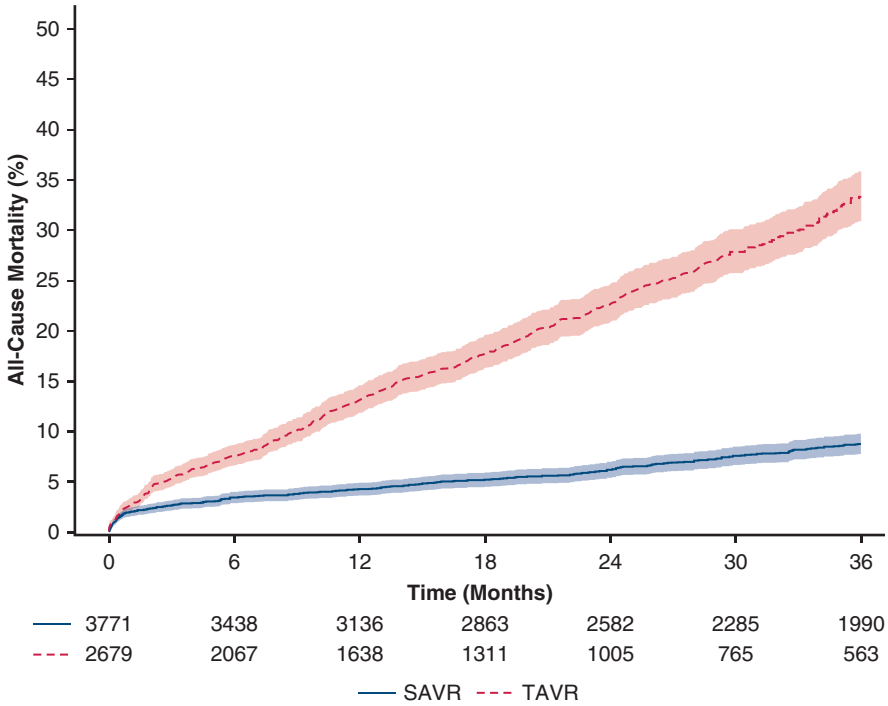

FIGURE E3. All-cause mortality (with 95% CI) after isolated SAVR versus TAVR for bicuspid AS in unmatched patients. *SAVR*, Surgical aortic valve replacement; *TAVR*, transcatheter aortic valve replacement.

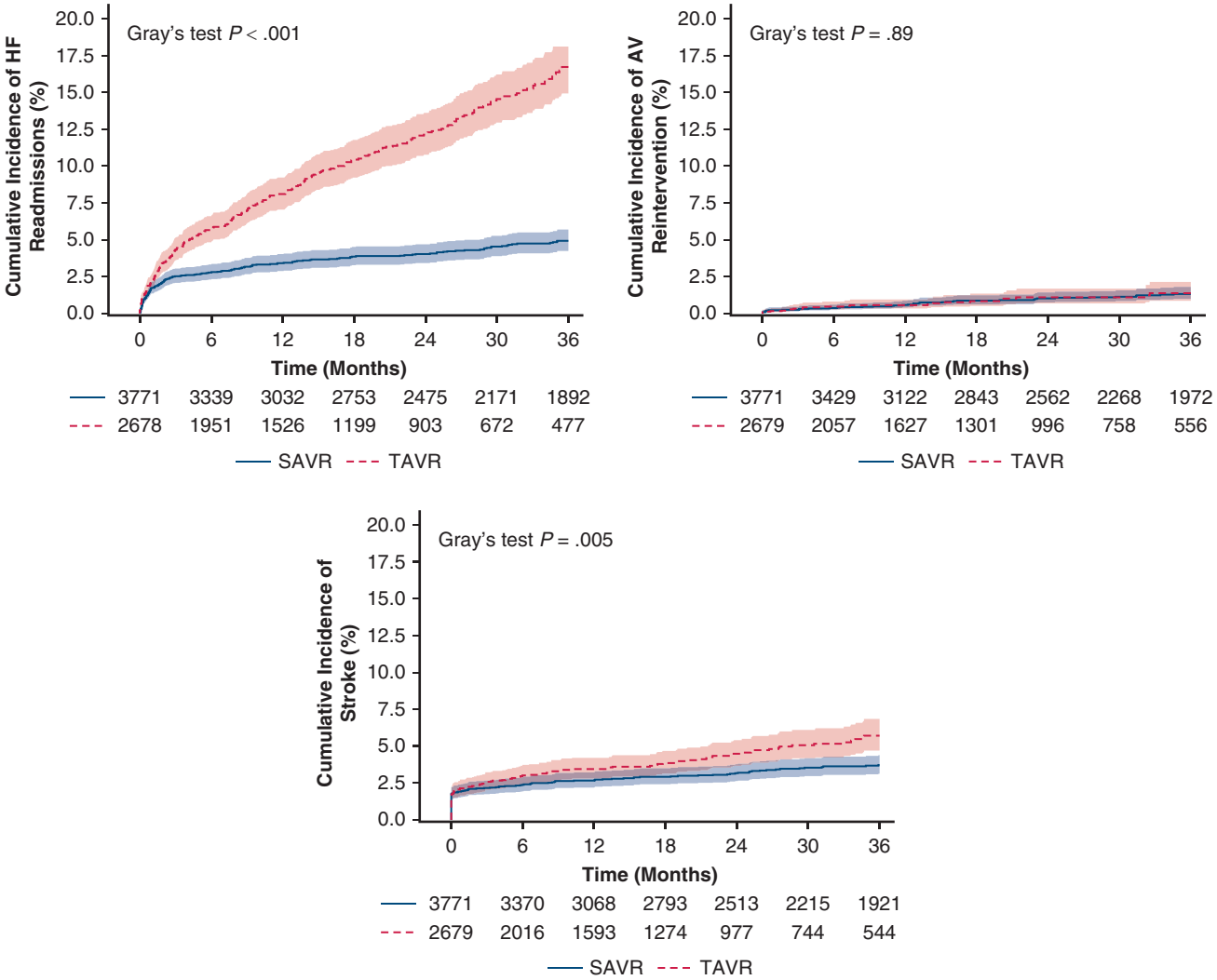

FIGURE E4. Secondary outcomes (with 95% CI) after isolated SAVR versus TAVR for bicuspid AS in unmatched patients. *HF*, Heart failure; *SAVR*, surgical aortic valve replacement; *TAVR*, transcatheter aortic valve replacement.

TABLE E1. International Classification of Diseases, 9th and 10th Revision codes for cohort identification

Diagnosis/procedure	ICD codes
Surgical aortic valve replacement	ICD-9-PCS: 3521, 3522 ICD-10-PCS: 02RF07Z, 02RF08Z, 02RF0KZ, 02RF0JZ, X2RF032
Transcatheter aortic valve replacement	ICD-9-PCS: 3505, 3506 ICD-10-PCS: 02RF37H, 02RF37Z, 02RF38H, 02RF38Z, 02RF3JH, 02RF3JZ, 02RF3KH, 02RF3KZ, X2RF332
Aortic stenosis	ICD-9-DX: 3950, 3960, 3962, 4241, 3952 ICD-10-DX: I060, I350, I352, I062
Bicuspid aortic valve	ICD-9-DX: 7463, 7464 ICD-10-DX: Q230, Q231
Concomitant valve procedures	Mitral valve procedures ICD-9-PCS: 3512, 3533, 3597, 3523, 3524 ICD-10-PCS: 02RG07Z, 02RG08Z, 02RG0JZ, 02RG0KZ, 02RG37H, 02RG37Z, 02RG38H, 02RG38Z, 02RG3JH, 02RG3JZ, 02RG3KH, 02RG3KZ, 02RG47Z, 02RG48Z, 02RG4JZ, 02RG4KZ, 027G04Z, 027G0DZ, 027G0ZZ, 027G34Z, 027G3DZ, 027G3ZZ, 027G44Z, 027G4DZ, 027G4ZZ,02QG0ZE, 02QG0ZZ, 02QG3ZE, 02QG3ZZ, 02QG4ZE,02QG4ZZ,02UG07Z,02UG08Z,02UG0JZ,02UG0KZ, 02UG37Z, 02UG38Z,02UG3JH,02UG3JZ,02UG3KZ, 02UG47Z, 02UG48Z, 02UG4JZ, 02UG4KZ, 02VG0ZZ,02WG07Z, 02WG08Z, 02WG0JZ, 02WG0KZ, 02WG37Z, 02WG38Z, 02WG3JZ, 02WG3KZ, 02WG47Z, 02WG48Z, 02WG4JZ, 02WG4KZ, 025G0ZZ, 025G3ZZ, 025G4ZZ, 02BG0ZZ, 02BG0ZZ, 02BG3ZZ, 02BG4ZZ, 02CG0ZZ, 02CG0ZZ, 02CG3ZZ, 02CG4ZZ, 02NG0ZZ, 02NG3ZZ, 02NG4ZZ, 02VG3ZZ, 02VG4ZZ Tricuspid valve procedures ICD-9-PCS: 3514, 3527, 3528, 3504 ICD-10-PCS: 02RJ07Z, 02RJ08Z, 02RJ0JZ, 02RJ0KZ, 02RJ37H, 02RJ37Z, 02RJ38H, 02RJ38Z, 02RJ3JH, 02RJ3JZ, 02RJ3KH, 02RJ3KZ, 02RJ47Z, 02RJ48Z, 02RJ4JZ, 02RJ4KZ, 027J0DZ, 027J0ZZ, 027J34Z, 027J44Z, 02QJ0ZG, 02QJ0ZZ, 02QJ3ZG, 02QJ4ZG, 02WJ07Z, 02WJ08Z, 02WJ0JZ, 02WJ0KZ, 02WJ37Z, 02WJ38Z, 02WJ3JZ, 02WJ3KZ, 02WJ47Z, 02WJ48Z, 02WJ4JZ, 02WJ4KZ, 025J0ZZ, 025J3ZZ, 025J4ZZ, 027J3DZ, 027J3ZZ, 027J4DZ, 027J4ZZ, 02BJ0ZZ, 02BJ3ZZ, 02BJ4ZZ, 02CJ0ZZ, 02CJ3ZZ, 02CJ4ZZ, 02NJ0ZZ, 02NJ0ZZ, 02NJ3ZZ, 02NJ3ZZ, 02NJ4ZZ, 02NJ4ZZ, 02QJ3ZZ, 02QJ4ZZ, 027J04Z, 02UJ07G, 02UJ07Z, 02UJ08, 02UJ08G, 02UJ08Z, 02UJ0J, 02UJ0JG, 02UJ0JZ,027J04Z, 02UJ07Z, 02UJ08Z, 02UJ0JZ, 02UJ0KZ, 02UJ37Z, 02UJ38Z, 02UJ3JZ, 02UJ3KZ, 02UJ47Z, 02UJ48Z, 02UJ4JZ, 02UJ4KZ Pulmonic valve and other unspecified valve procedures ICD-9-PCS: 3525, 3526, 3508, 3513, 3507, 3508, 3509, 3510, 3511, 3595, 3596, 3599 ICD-10-PCS: 02RH07Z, 02RH08Z, 02RH0JZ, 02RH0KZ, 02RH37H, 02RH37Z, 02RH38H, 02RH38Z, 02RH3JH, 02RH3JZ, 02RH3KH, 02RH3KZ, 02RH47Z, 02RH48Z, 02RH4JZ, 02RH4KZ, 027H04Z, 027H0DZ, 027H0ZZ, 027H34Z, 027H44Z, 02QH0ZZ, 02UH07Z, 02UH08Z, 02UH0JZ, 02UH0KZ, 02UH37Z, 02UH38Z, 02UH3JZ, 02UH3KZ, 02UH47Z, 02UH48Z, 02UH4JZ, 02UH4KZ, 02WH07Z, 02WH08Z, 02WH0JZ, 02WH0KZ, 02WH37Z, 02WH38Z, 02WH3JZ, 02WH3KZ, 02WH47Z, 02WH48Z, 02WH4JZ, 02WH4KZ, 02NH0ZZ Coronary artery bypass grafting ICD-9-PCS: 361, 3610, 3611, 3612, 3613, 3614, 3615, 3616, 3619, 3631, 3632, 3639, 362, 3717 ICD-10-PCS: 0210093, 0210098, 0210099, 021009C, 021009F, 021009W, 02100A3, 02100A8, 02100A9, 02100AC, 02100AF, 02100AW, 02100J3, 02100J8, 02100J9, 02100JC, 02100JF, 02100JW, 02100K3, 02100K8, 02100K9, 02100KC, 02100KF, 02100KW, 02100Z3, 02100Z8, 02100Z9, 02100ZC, 02100ZF, 0210493, 0210498, 0210499, 021049C, 021049F, 021049W, 02104A3, 02104A8, 02104A9, 02104AC, 02104AF, 02104AW, 02104J3, 02104J8, 02104J9, 02104JC, 02104JF, 02104JW, 02104K3, 02104K8, 02104K9, 02104KC, 02104KF, 02104KW, 02104Z3, 02104Z8, 02104Z9, 02104ZC, 02104ZF, 0211093, 0211098, 0211099, 021109C, 021109F, 021109W, 02110A3, 02110A8, 02110A9, 02110AC, 02110AF, 02110AW, 02110J3, 02110J8, 02110J9, 02110JC, 02110JF, 02110JW, 02110K3, 02110K8, 02110K9, 02110KC, 02110KF, 02110KW, 02110Z3, 02110Z8, 02110Z9, 02110ZC, 02110ZF, 0211493, 0211498, 0211499, 021149C, 021149F, 021149W, 02114A3, 02114A8, 02114A9, 02114AC, 02114AF, 02114AW, 02114J3, 02114J8, 02114J9, 02114JC, 02114JF, 02114JW, 02114K3, 02114K8, 02114K9, 02114KC, 02114KF, 02114KW, 02114Z3, 02114Z8, 02114Z9, 02114ZC, 02114ZF, 0212093, 0212098, 0212099, 021209C, 021209F, 021209W, 02120A3, 02120A8, 02120A9, 02120AC, 02120AF, 02120AW, 02120J3, 02120J8, 02120J9, 02120JC, 02120JF, 02120JW, 02120K3, 02120K8, 02120K9, 02120KC, 02120KF, 02120KW, 02120Z3, 02120Z8, 02120Z9, 02120ZC, 02120ZF, 0212493, 0212498, 0212499, 021249C, 021249F, 021249W, 02124A3, 02124A8, 02124A9, 02124AC, 02124AF, 02124AW, 02124J3, 02124J8, 02124J9, 02124JC, 02124JF, 02124JW, 02124K3,

(Continued)

TABLE E1. Continued

Diagnosis/procedure	ICD codes
	02124K8, 02124K9, 02124KC, 02124KF, 02124KW, 02124Z3, 02124Z8, 02124Z9, 02124ZC, 02124ZF, 0213093, 0213098, 0213099, 021309C, 021309F, 021309W, 02130A3, 02130A8, 02130A9, 02130AC, 02130AF, 02130AW, 02130J3, 02130J8, 02130J9, 02130JC, 02130JF, 02130JW, 02130K3, 02130K8, 02130K9, 02130KC , 02130KF, 02130KW, 02130Z3, 02130Z8, 02130Z9, 02130ZC, 02130ZF, 0213493, 0213498, 0213499, 021349C, 021349F, 021349W, 02134A3, 02134A8, 02134A9, 02134AC, 02134AF, 02134AW, 02134J3, 02134J8, 02134J9, 02134JC, 02134JF, 02134JW, 02134K3, 02134K8, 02134K9, 02134KC, 02134KF, 02134KW, 02134Z3, 02134Z8, 02134Z9, 02134ZC, 02134ZF, 021008W, 021048W, 021148W, 021308W, 021348W, 0210088, 0210089, 021008C, 0210488, 0210489, 021048C, 0211083, 0211088 0211089, 021108C, 021108F, 021108W, 0211483, 0211488, 0211489, 021148C, 021148F, 0212083, 0212088, 0212089, 021208C, 021208F, 021208W, 0212483, 0212488, 0212489, 021248C, 021248F, 021248W, 0213083, 0213088, 0213089, 021308C, 021308F, 0213483, 0213488, 0213489, 021348C, 021348F, 021008F, 021048F, 0210083, 0210483
	Percutaneous coronary interventions
	ICD-9-PCS: 0066, 1755, 3601, 3602, 3603, 3605, 3606, 3607, 3609, 3633, 3634
	ICD-10-PCS: 0270046, 027004Z, 0270056, 027005Z, 0270066, 027006Z, 0270076, 027007Z, 02700D6, 02700DZ, 02700E6, 02700EZ, 02700F6, 02700FZ, 02700G6, 02700GZ, 02700Z6, 02700ZZ, 0270346, 027034Z, 0270356, 027035Z, 0270366, 027036Z, 0270376, 027037Z, 02703D6, 02703DZ, 02703E6, 02703EZ, 02703F6, 02703FZ, 02703G6, 02703GZ, 02703Z6, 02703ZZ, 0270446, 027044Z, 0270456, 027045Z, 0270466, 027046Z, 0270476, 027047Z, 02704D6, 02704DZ, 02704E6, 02704EZ, 02704F6, 02704FZ, 02704G6, 02704GZ, 02704Z6, 02704ZZ, 0271046, 027104Z, 0271056, 027105Z, 0271066, 027106Z, 0271076, 027107Z, 02710D6, 02710DZ, 02710E6, 02710EZ, 02710F6, 02710FZ, 02710G6, 02710GZ, 02710Z6, 02710ZZ, 0271346, 027134Z, 0271356, 027135Z, 0271366, 027136Z, 0271376, 027137Z, 02713D6, 02713D6, 02713E6, 02713EZ, 02713F6, 02713FZ, 02713G6, 02713GZ, 02713Z6, 02713ZZ, 0271446, 027144Z, 0271456, 027145Z, 0271466, 027146Z, 0271476, 027147Z, 02714D6, 02714DZ, 02714E6, 02714EZ, 02714F6, 02714FZ, 02714G6, 02714GZ, 02714Z6, 02714ZZ, 0272046, 027204Z, 0272056, 027205Z, 0272066, 027206Z, 0272076, 027207Z, 02720D6, 02720DZ, 02720E6, 02720EZ, 02720F6, 02720FZ, 02720G6, 02720GZ, 02720Z6, 02720ZZ, 0272346, 027234Z, 0272356, 027235Z, 0272366, 027236Z, 0272376, 027237Z, 02723D6, 02723DZ, 02723E6, 02723EZ, 02723F6, 02723FZ, 02723G6, 02723GZ, 02723Z6, 02723ZZ, 0272446, 027244Z, 0272456, 027245Z, 0272466, 027246Z, 0272476, 027247Z, 02724D6, 02724DZ, 02724E6, 02724EZ, 02724F6, 02724FZ, 02724G6, 02724GZ, 02724Z6, 02724ZZ, 0273046, 027304Z, 0273056, 027305Z, 0273066, 027306Z, 0273076, 027307Z, 02730D6, 02730DZ, 02730E6, 02730EZ, 02730F6, 02730FZ, 02730G6, 02730GZ, 02730Z6, 02730ZZ, 0273346, 027334Z, 0273356, 027335Z, 0273366, 027336Z, 0273376, 027337Z, 02733D6, 02733DZ, 02733E6, 02733EZ, 02733F6, 02733FZ, 02733G6, 02733GZ, 02733Z6, 02733ZZ, 0273446, 027344Z, 0273456, 027345Z, 0273466, 027346Z, 0273476, 027347Z, 02734D6, 02734DZ, 02734E6, 02734EZ, 02734F6, 02734FZ, 02734G6, 02734GZ, 02734Z6, 02734ZZ, 02C00Z6, 02C00ZZ, 02C03Z6, 02C03ZZ, 02C04Z6, 02C04ZZ, 02C10Z6, 02C10ZZ, 02C13Z6, 02C13ZZ, 02C14Z6, 02C14ZZ, 02C20Z6, 02C20ZZ, 02C23Z6, 02C23ZZ, 02C24Z6, 02C24ZZ, 02C30Z6, 02C30ZZ, 02C33Z6, 02C33ZZ, 02C34Z6, 02C34ZZ, 3E07017, 3E070PZ, 3E07317, 3E073PZ, 02H03DZ, 02H03YZ, 02H13DZ, 02H13YZ, 02H23DZ, 02H23YZ,
	02H33DZ, 02H33YZ, 02H04DZ, 02H04YZ, 02H14DZ, 02H14YZ, 02H24DZ, 02H24YZ, 02H34DZ, 02H34YZ, X2C0361, X2C1361, X2C2361, X2C3361
	Ventricular assist device implantation or heart transplantation
	ICD-9-PCS: 336, 3751, 3752, 375, 3753, 3754, 3755, 3760, 3762, 3763, 3765, 3766, 3768
	ICD-10-PCS: 02HA0QZ, 02HA0RJ, 02HA0RS, 02HA0RZ, 02HA3QZ, 02HA3RJ, 02HA3RS, 02HA3RZ, 02HA4QZ, 02HA4RJ, 02HA4RS, 02HA4RZ, 02RK07Z, 02RK08Z, 02RK0JZ, 02RK0KZ, 02RK47, 02RK47Z, 02RK48Z, 02RK4JZ, 02RK4KZ, 02RL07Z, 02RL08Z, 02RL0JZ, 02RL0KZ, 02RL47Z, 02RL48Z, 02RL4JZ, 02RL4KZ, 02UA0JZ, 02UA3JZ, 02UA4JZ, 02WA0JZ, 02WA0QZ, 02WA3QZ, 02YA0Z0, 02YA0Z1, 02YA0Z2, 5A02116, 5A0211D
	Procedures on the thoracic aorta
	ICD-9-PCS: 3593, 3804, 3805, 3814, 3834, 3845, 3865, 3922, 3923, 3954, 3978, 3979
	ICD-10-PCS: 021K0ZW, 021L0ZW, 021L4ZW, 021W08A, 021W08B, 021W08D, 021W08F, 021W08G, 021W08P, 021W08Q, 021W08R, 021W09A, 021W09B, 021W09D, 021W09F, 021W09G, 021W09P, 021W09Q, 021W09R, 021W0AA, 021W0AB, 021W0AD, 021W0AF, 021W0AG, 021W0AP, 021W0AQ, 021W0AR, 021W0JA, 021W0JB, 021W0JD, 021W0JF, 021W0JG, 021W0JP, 021W0JQ, 021W0JR, 021W0KA, 021W0KB, 021W0KD, 021W0KF, 021W0KG, 021W0KP, 021W0KQ, 021W0KR, 021W0ZA, 021W0ZB, 021W0ZD, 021W0ZP, 021W0ZQ, 021W0ZR, 021W48A, 021W48B, 021W48D, 021W48P, 021W48Q, 021W48R, 021W49A, 021W49B, 021W49D, 021W49P, 021W49Q, 021W49R, 021W4AA, 021W4AB, 021W4AD, 021W4AP, 021W4AQ,

(Continued)

TABLE E1. Continued

Diagnosis/procedure	ICD codes
	021W4AR, 021W4JA, 021W4JB, 021W4JD, 021W4JP, 021W4JQ, 021W4JR, 021W4KA, 021W4KB, 021W4KD, 021W4KP, 021W4KQ, 021W4KR, 021W4ZA, 021W4ZB, 021W4ZD, 021W4ZP, 021W4ZQ, 021W4ZR, 021X08A, 021X08A, 021X08B, 021X08D, 021X08P, 021X08Q, 021X08R, 021X09A, 021X09B, 021X09D, 021X09Q, 021X09R, 021X0AA, 021X0AB, 021X0AD, 021X0AP, 021X0AQ, 021X0AR, 021X0JA, 021X0JB, 021X0JD, 021X0JP, 021X0JQ, 021X0JR, 021X0KA, 021X0KB, 021X0KD, 021X0KP, 021X0KQ, 021X0KR, 021X0ZA, 021X0ZB, 021X0ZD, 021X0ZP, 021X0ZQ, 021X0ZR, 021X48A, 021X48B, 021X48D, 021X48P, 021X48Q, 021X48R, 021X49A, 021X49B, 021X49D, 021X49P, 021X49Q, 021X49R, 021X4AA, 021X4AB, 021X4AD, 021X4AP, 021X4AQ, 021X4AR, 021X4JA, 021X4JB, 021X4JD, 021X4JP, 021X4JQ, 021X4JR, 021X4KA, 021X4KB, 021X4KD, 021X4KP, 021X4KQ, 021X4KR, 021X4ZA, 021X4ZB, 021X4ZD, 021X4ZP, 021X4ZQ, 021X4ZR, 025W0ZZ, 025W3ZZ, 025W4ZZ, 025X0ZZ, 025X3ZZ, 025X4ZZ, 02BW0ZZ, 02BW3ZZ, 02BW4ZZ, 02BX0ZZ, 02BX3ZZ, 02BX4ZZ, 02CW0ZZ, 02CW3ZZ, 02CW4ZZ, 02CX0ZZ, 02CX3ZZ, 02CX4ZZ, 02HW0DZ, 02HW0YZ, 02HW3DZ, 02HW3YZ, 02HW4DZ, 02HW4YZ, 02HX0DZ, 02HX3DZ, 02HX4DZ, 02NW0ZZ, 02NW3ZZ, 02NW4ZZ, 02NX0ZZ, 02NX3ZZ, 02NX4ZZ, 02QW0ZZ, 02QW3ZZ, 02QW4ZZ, 02QX0ZZ, 02QX3ZZ, 02QX4ZZ, 02RW07Z, 02RW08Z, 02RW0JZ, 02RW0KZ, 02RW47Z, 02RW48Z, 02RW4JZ, 02RW4KZ, 02RX07Z, 02RX08Z, 02RX0JZ, 02RX0KZ, 02RX47Z, 02RX48Z, 02RX4JZ, 02RX4KZ, 02SW0ZZ, 02SX0ZZ, 02UW07Z, 02UW08Z, 02UW0JZ, 02UW0KZ, 02UW37Z, 02UW38Z, 02UW3JZ, 02UW3KZ, 02UW47Z, 02UW48Z, 02UW4JZ, 02UW4KZ, 02UX07Z, 02UX08Z, 02UX0JZ, 02UX0KZ, 02UX37Z, 02UX38Z, 02UX3JZ, 02UX3KZ, 02UX47Z, 02UX48Z, 02UX4JZ, 02UX4KZ, 02VW0DZ, 02VW0EZ, 02VW0FZ, 02VW0ZZ, 02VW3CZ, 02VW3DZ, 02VW3EZ, 02VW3FZ, 02VW3ZZ, 02VW4CZ, 02VW4DZ, 02VW4EZ, 02VW4FZ, 02VW4ZZ, 02VX0CZ, 02VX0DZ, 02VX0EZ, 02VX0FZ, 02VX0ZZ, 02VX3CZ, 02VX3DZ, 02VX3EZ, 02VX3FZ, 02VX3ZZ, 02VX4CZ, 02VX4DZ, 02VX4EZ, 02VX4FZ, 02VX4ZZ, 02VW0CZ

ICD-9, International Classification of Diseases, 9th Revision; *ICD-10*, International Classification of Diseases, 10th Revision; *PCS*, procedure codes; *DX*, diagnosis codes.

TABLE E2. Definition of baseline comorbidities

Baseline comorbidities	ICD codes from index and prior admissions
Hypertension	ICD-9-DX: 401, 4010, 4011, 4019, 402, 40200, 40201, 4021, 40210, 40211, 4029, 40290, 40291, 403, 4030, 40300, 40301, 4031, 40310, 40311, 4039, 40390, 40391, 404, 4040, 40400, 40401, 40402, 40403, 4041, 40410, 40411, 40412, 40413, 4049, 40490, 40491, 40492, 40493, 405, 4050, 40501, 40509, 4051, 40511, 40519, 4059, 40591, 40599, 4372 ICD-10-DX: I10, I110, I119, I129, I120, I1310, I130, I1311, I132, I150, I158, I674
Hyperlipidemia	ICD-9-DX: 2724 ICD-10-DX: E784, E7841, E7849, E785
Diabetes	ICD-9-DX: 249, 2490, 24900, 24901, 2491, 24910, 24911, 2492, 24920, 24921, 2493, 24930, 24931, 2494, 24940, 24941, 2495, 24950, 24951, 2496, 24960, 24961, 2497, 24970, 24971, 2498, 24980, 24981, 2499, 24990, 24991, 250, 2500, 25000, 25001, 25002, 25003, 2501, 25010, 25011, 25012, 25013, 2502, 25020, 25021, 25022, 25023, 2503, 25030, 25031, 25032, 25033, 2504, 25040, 25041, 25042, 25043, 2505, 25050, 25051, 25052, 25053, 2506, 25060, 25061, 25062, 25063, 2507, 25070, 25071, 25072, 25073, 2508, 25080, 25081, 25082, 25083, 2509, 25090, 25091, 25092, 25093 ICD-10-DX: E0801, E0810, E0811, E0821, E08311, E08319, E0836, E0839, E0840, E0841, E0842, E0843, E0844, E0849, E0851, E08610, E08618, E08620, E08621, E08622, E08628, E08630, E08638, E08641, E0865, E0869, E088, E089, E0901E0910, E0911, E0921, E09311, E09319, E0936, E0939, E0940, E0941, E0942, E0943, E0944, E0949, E0951, E09610, E09618, E09620, E09621, E09622, E09628, E09630, E09638, E09641, E09649, E0965, E0969, E098, E099, E1021, E1029, E10311, E10319, E1036, E1039, E1040, E1051, E1065, E108, E109, E1121, E1129, E11311, E11319, E113499, E11351, E113511, E113512, E113513, E113519, E113521, E113522, E113523, E113529, E113531, E113532, E113533, E113539, E113541, E113542, E113543, E113549, E113551, E113552, E113553, E113559, E11359, E113591, E113592, E113593, E113599, E1136, E1137X1, E1137X2, E1137X3, E1137X9, E1139, E1140, E1141, E1142, E1143, E1144, E1149, E1151, E1152, E1159, E11610, E11618, E11620, E11621, E11622, E11628, E11630, E11638, E11641, E11649, E1165, E1169, E118, E119, E1300, E1301, E1310, E1311, E1321, E1322, E1329, E13311, E13319, E13321, E133211, E133212, E133213, E133219, E13329, E133291, E133292, E133293, E133299, E13331, E133311, E133312, E133313, E133319, E13339, E133391, E133392, E133393, E133399, E13341, E133411, E133412, E133413, E133419, E13349, E133491, E133492, E133493, E133499, E13351, E133511, E133512, E133513, E133519, E133521, E133522, E133523, E133529, E133531, E133532, E133533, E133539, E133541, E133542, E133543, E133549, E133551, E133552, E133553, E133559, E13359, E133591, E133592, E133593, E133599, E1336, E1339, E1340, E1341, E1342, E1343, E1344, E1349, E1351, E1352, E1359, E13610, E13618, E13620, E13621, E13622, E13628, E13630, E13638, E13641, E13649, E1365, E1369, E138, E139
Peripheral vascular disease	ICD-9-DX: 4400, 4401, 4402, 44020, 44021, 44022, 44023, 44024, 44029, 4403, 44030, 44031, 44032, 4404, 4408, 4409, 4471, 9961, 99662, 99674 and V434 ICD-10-DX: E0851, E0852, E0951, E0952, E1051, E1052, E1151, E1152, E1159, E1351, E1352, E1359, I700, I701, I70201, I70202, I70203, I70208, I70209, I70211, I70212, I70213, I70218, I70219, I70221, I70222, I70223, I70228, I70229, I70231, I70232, I70233, I70234, I70235, I70238, I70239, I70241, I70242, I70243, I70244, I70245, I70248, I70249, I7025, I70269, I70291, I70292, I70293, I70298, I70299, I70399, I70499, I70599, I708, I7090, I7091, I7092, I731, I7389, I739, I791, I798
Prior stroke	ICD-9-DX: 99702, 431, 432, 4320, 4321, 4329, 43311, 43321, 43331, 43381, 43391, 43401, 43411, 43491, I97811, I97821, 430, 43301 ICD-10-DX: 19781, 19782, 197810, 197811, 197820, 197821, I60, I600, I6000, I6001, I6002, I601, I6010, I6011, I6012, I602, I603, I6030, I6031, I6032, I605, I6050, I6051, I6052, I606, I607, I608, I609, I61, I610, I611, I612, I613, I614, I615, I604, I616, G460, G461, G462, G463, G464, G465, G466, G467, I63, I630, I6300, I6301, I63011, I63012, I63013, I63019, I6302, I6303, I63031, I63032, I63033, I63039, I6309, I631, I6310, I6311, I63111, I63112, I63113, I63119, I6312, I6313, I63131, I63132, I63133, I63139, I6319, I632, I6320, I6321, I63211, I63212, I63213, I63219, I6322, I6323, I63231, I63232, I63233, I63239, I6329, I633, I6330, I6331, I63311, I63312, I63313, I63319, I6332, I63321, I63322, I63323, I63329, I6333, I63331, I63332, I63333, I63339, I6334, I63341, I63342, I63343, I63349, I6339, I634, I6340, I6341, I63411, I63412, I63413, I63419, I6342, I63421, I63422, I63423, I63429, I6343, I63431, I63432, I63433, I63439, I6344, I63441, I63442, I63443, I63449, I6349, I635, I6350, I6351, I63511, I63512, I63513, I63519, I6352, I63521, I63522, I63523, I63529, I6353, I63531, I63532, I63533, I63539, I6354, I63541, I63542, I63543, I63549, I6359, I636, I638, I6381, I6389, I639, I6781, I6782, I6789

(Continued)

TABLE E2. Continued

Baseline comorbidities	ICD codes from index and prior admissions
Cerebrovascular disease	ICD-9-DX: 3623, 36230, 36231, 36232, 36233, 36234, 36235, 36236, 36237, 3466, 34660, 34661, 34662, 34663, 430, 431, 432, 4320, 4321, 4329, 433, 4330, 43300, 43301, 4331, 43310, 43311, 4332, 43320, 43321, 4333, 43330, 43331, 4338, 43380, 43381, 4339, 43390, 43391, 434, 4340, 43400, 43401, 4341, 43410, 43411, 4349, 43490, 43491, 435, 4350, 4351, 4352, 4353, 4358, 4359, 436, 437, 4370, 4371, 4372, 4373, 4374, 4375, 4376, 4377, 4378, 4379, 438, 4380, 4381, 43810, 43811, 43812, 43813, 43814, 43819, 4382, 43820, 43821, 43822, 4383, 43830, 43831, 43832, 4384, 43840, 43841, 43842, 4385, 43850, 43851, 43852, 4386, 4387, 4388, 43881, 43882, 43883, 43884, 43885, 43889, 4389 ICD-10-DX: 19781, 19782, 197810, 197820, G43601, G43611, G43619, G450, G451, G454, G458, G459, G460, G461, G462, G463, G464, G465, G466, G467, H3400, H3413, H34219, H34239, H348110, H348111, H348112, H348120, H348121, H348122, H348130, H348131, H348132, H348190, H348191, H348192, H34829, H348310, H348311, H348312, H348320, H348321, H348322, H348330, H348331, H348332, H348390, H348391, H348392, H349, I63, I630, I6300, I6301, I63011, I63012, I63013, I63019, I6302, I6303, I63031, I63032, I63033, I63039, I6309, I631, I6310, I6311, I63111, I63112, I63113, I63119, I6312, I6313, I63131, I63132, I63133, I63139, I6319, I632, I6320, I6321, I63211, I63212, I63213, I63219, I6322, I6323, I63231, I63232, I63233, I63239, I6329, I633, I6330, I6331, I63311, I63312, I63313, I63319, I6332, I63321, I63322, I63323, I63329, I6333, I63331, I63332, I63333, I63339, I6334, I63341, I63342, I63343, I63349, I6339, I634, I6340, I6341, I63411, I63412, I63413, I63419, I6342, I63421, I63422, I63423, I63429, I6343, I63431, I63432, I63433, I63439, I6344, I63441, I63442, I63443, I63449, I6349, I635, I6350, I6351, I63511, I63512, I63513, I63519, I6352, I63521, I63522, I63523, I63529, I6353, I63531, I63532, I63533, I63539, I6354, I63541, I63542, I63543, I63549, I6359, I636, I638, I6381, I6389, I639, I65, I650, I6501, I6502, I6503, I6509, I651, I652, I6521, I6522, I6523, I6529, I658, I659, I66, I660, I6601, I6602, I6603, I6609, I661, I6611, I6612, I6613, I6619, I662, I6621, I6622, I6623, I6629, I663, I668, I669, I67, I670, I671, I672, I673, I674, I675, I676, I677, I678, I6781, I6782, I6783, I6784, I67841, I67848, I6785, I67850, I67858, I6789, I679, I68, I680, I682, I688, I69, I69898, I6990, I69910, I69911, I69912, I69913, I69914, I69915, I69918, I69919, I69920, I69921, I69922, I69923, I69928, I69931, I69932, I69933, I69934, I69943, I69944, I69951, I69952, I69953, I69954, I69961, I69962, I69963, I69964, I69990, I69991, I69992, I69993, I69998, I97811, I97821
Coronary artery disease	ICD-9-DX: 410, 4100, 41000, 41001, 41002, 4101, 41010, 41011, 41012, 4012, 40120, 41021, 41022, 4103, 41030, 41031, 41032, 4104, 41040, 41041, 41042, 4105, 41050, 41051, 41052, 4106, 41060, 41061, 41062, 4107, 41070, 41071, 41072, 4108, 41080, 41081, 41082, 4109, 41090, 41091, 41092, 411, 4110, 4111, 4118, 41181, 41189, 412, 413, 4130, 4131, 4139, 414, 4140, 41400, 41401, 41402, 41403, 41404, 41405, 41406, 41407, 4142, 4143, 4144, 4148, 4149, 4295, 4296, 4297, 42971, 42979 ICD-10-DX: I2109, I2119, I2111, I2129, I214, I213, I241, I200, I240, I248, I252, I208, I201, I2510, I25810, I512, I510, I230, I209, I25, I251, I2511, I25110, I25111, I25118, I25119
Congestive heart failure	ICD-9-DX: 39891, 428, 4280, 4281, 4282, 42820, 42821, 42822, 42823, 4283, 42830, 42831, 42832, 42833, 4284, 42840, 42841, 42842, 42843, 4289, 429, 4290, 4291, 4292, 4293, 4294, 4295, 4296, 4297, 42971, 42979, 4298, 42981, 42982, 42983, 42989, 4299 ICD-10-DX: ICD10: I0981, I110, I130, I132, I501, I5020, I5021, I5022, I5023, I5030, I5031, I5032, I5033, I5040, I5041, I5042, I5043, I50810, I50811, I50812, I50813, I50814, I5082, I5083, I5084, I5089, I509 , I5181, I97110, I97130, I97190
Atrial fibrillation	ICD-9-DX: 4273, 42731, 42732 ICD-10-DX: I483, I484, I4892, I480, I481, I4811, I4819, I482, I4820, I4821, I4891, I4892
Chronic obstructive pulmonary disease	ICD-9-DX: 491, 4910, 4911, 4912, 49120, 49121, 49122, 4918, 4919, 492, 4920, 4928, 493, 4930, 49300, 49301, 49302, 4931, 49310, 49311, 49312, 4932, 49320, 49321, 49322, 4938, 49380, 49381, 49382, 4939, 49390, 49391, 49392, 494, 4940, 4941, 496 ICD-10-DX: J410, J411, J418, J42, J430, J431, J432, J438, J439, J440, J441, J449, J470 , J471, J479, J982, J983
Chronic kidney disease	ICD-9-DX: 403, 404, 556, 582, 583, 585, 586, 587, 588, 2494, 2504, 4030, 4031, 4039, 4040, 4041, 4049, 5569, 5820, 5821, 5822, 5824, 5828, 5829, 5830, 5831, 5832, 5834, 5836, 5837, 5838, 5839, 5851, 5852, 5853, 5854, 5855, 5859, 5880, 5881, 5888, 5889, 24940, 24941, 25040, 25041, 25042, 25043, 40300, 40301, 40310, 40311, 40390, 40391, 40400, 40401, 40402, 40403, 40410, 40411, 40412, 40413, 40490, 40491, 40492, 40493, 58281, 58289, 58381, 58389, 58881, 58889, 99681, V420 ICD-10-DX: E0821, E0822, E0829, E0921, E0922, E0929, E1021, E1022, E1029, E1121, E1122, E1129, E1321, E1322, E1329, I12, I129, I130, I131, I1310, I1311, I132, M1030, M10311, M10312, M10319,

(Continued)

TABLE E2. Continued

Baseline comorbidities	ICD codes from index and prior admissions
	M10321, M10322, M10329, M10331, M10332, M10339, M10341, M10342, M10349, M10351, M10352, M10359, M10361, M10362, M10369, M10371, M10372, M10379, M1038, M1039, M3214, M3215, M3504, N03, N030, N031, N032, N033, N034, N035, N036, N037, N038, N039, N03A, N051, N052, N053, N054, N055, N056, N057, N058, N059, N05A, N08, N110, N111, N118, N119, N12, N181, N182, N183, N1830, N1831, N1832, N184, N185, N189, N19, N25, N250, N251, N2581, N2589, N259, N261, N269, N990, Z490, Z493, Z940
Dialysis dependency	ICD-9-DX: 7925, V511, V563, V5631, V5632, V568, E8791, V451, V4511, V560, V561, V562, 99656, 99668, E8702, E8712, E8722, E8742 ICD-9-PCS: 3895, 3927, 3942, 3943, 3995, 5494, 5498 ICD-10-DX: N186, I120, I1311, I132, Z4092, Z4901, Z4931, Z4932, Z992, Y841, Z9115, T824, Y622, Z49, Z490, 99673, V4512, T81502, T81512, T81522, T81532, T81592, T8241, T8249, T85611, T85621, T85691, Z491, Z492, Z493, T8571XA, T8571XD, T8571XS, T8242XA, T8242XD, T8242XS, T8243XA, T8243XD, T8243XS, T8249XA, T8249XD, T8249XS, T85631A, T85631D, T85631S, T85621A, T85621D, T85621S, T85611A, T85611D, T85611S, T85691A, T85691D, T85691S, T8249XA, T8249XD, T8249XS, T81512A, T81512D, T81512S, T81522A, T81522D, T81522S, T81532A, T81532S, T81532D, T81592A, T81592S, T81532D, T8241XA, T8241XD, T8241XS, T8242XA, T8242XD, T8242XS, T8243XD, T8243XA, T8243XS, T82510A, T82511A, T82520A, T82521A, T82530A, T82531A, R880, T8241, T8242, T8243, T8249, T856315 ICD-10-PCS: 0W1G0JY, 0W1G4JY, 5A1D70Z, 5A1D80Z, 5A1D90Z, B50W, B50W0ZZ, B50W1ZZ, B50WYZZ, B51W, B51W0ZA, B51W0ZZ, B51W1ZA, B51W1ZZ, B51WYZA, B51WYZZ, B51WZZA, B51WZZZ
Liver disease	ICD-9-DX: 070, 0700, 0701, 0702, 07020, 07021, 07022, 07023, 0703, 07030, 07031, 07032, 07033, 0704, 07041, 07042, 07043, 07044, 07049, 0705, 07051, 07052, 07053, 07054, 07059, 0706, 0707, 07070, 07071, 0709, 456, 4560, 4561, 4562, 45620, 45621, 4563, 4564, 4565, 4566, 4568, 570, 571, 5710, 5711, 5712, 5713, 5714, 57140, 57141, 57142, 57149, 5715, 5716, 5718, 5719, 572, 5720, 5721, 5722, 5723, 5724, 5728, 573, 5730, 5731, 5732, 5733, 5734, 5735, 5738, 5738, 7824, 7891, 7895, 78951, 78959, 7904, 7905, 7948, V427 ICD-10-DX: B182, B188, B189, I8500, I8501, I8510, I8511, I860, I861, I862, I863, I868, K700, K7010, K7011, K702, K7030, K7031, K7040, K7041, K709, K710, K7111, K716, K717, K718, K719, K7200, K7201, K7210, K7211, K7290, K7291, K730, K732, K738, K739, K740, K7400, K7401, K7402, K741, K742, K743, K744, K745, K7460, K7469, K750, K751, K754, K7581, K7589, K759, K760, K761, K762, K763, K765, K766, K767, K7681, K7689, K769, K77, R160, R162, R17, R180, R188, Z944
Cancer (including oropharyngeal cancers, gastrointestinal cancers, respiratory tract cancers, bone and connective tissue cancer, genitourinary cancer, lymphoid cancer, hematological cancer, and other cancers)	ICD-9-DX: 140, 1400, 1401, 1403, 1404, 1405, 1406, 1408, 1409, 141, 1410, 1411, 1412, 1413, 1414, 1415, 1416, 1418, 1419, 142, 1420, 1421, 1422, 1428, 1429, 143, 1430, 1431, 1438, 1439, 144, 1440, 1441, 1448, 1449, 145, 1450, 1451, 1452, 1453, 1454, 1455, 1456, 1458, 1459, 146, 1460, 1461, 1462, 1463, 1464, 1465, 1466, 1467, 1468, 1469, 147, 1470, 1471, 1472, 1473, 1478, 1479, 148, 1480, 1481, 1482, 1483, 1488, 1489, 149, 1490, 1491, 1498, 1499, 150, 1500, 1501, 1502, 1503, 1504, 1505, 1506, 1508, 1509, 1510, 151, 1511, 15012, 1513, 1514, 1515, 1516, 1518, 1519, 152, 1520, 1521, 1522, 1523, 1528, 1529, 153, 1530, 1531, 1532, 1533, 1534, 1535, 1536, 1537, 1538, 1539, 154, 1540, 1541, 1542, 1543, 1548, 155, 1550, 1551, 1552, 156, 1560, 1561, 1562, 1568, 1569, 157, 1570, 1571, 1572, 1573, 1574, 1578, 1579, 158, 1580, 1588, 1589, 159, 1590, 1591, 1598, 1599, 160, 1600, 1601, 1602, 1603, 1604, 1605, 1608, 1609, 161, 1610, 1611, 1612, 1613, 1618, 1619, 162, 1620, 1622, 1623, 1624, 1625, 1628, 1629, 163, 1630, 1631, 1638, 1639, 164, 1640, 1461, 1462, 1463, 1468, 1649, 165, 1650, 1658 and 1659, 170, 1700, 1701, 1702, 1703, 1704, 1705, 1706, 1707, 1708, 1709, 171, 1710, 1712, 1713, 1714, 1715, 1716, 1717, 1718, 1719, 172, 1720, 1721, 1722, 1723, 1724, 1725, 1726, 1727, 1728, 1729, 173, 1730, 17300, 17301, 17302, 17309, 1731, 17310, 17311, 17312, 17319, 1732, 17320, 17321, 17322, 17329, 1733, 17330, 17331, 17332, 17339, 1734, 17340, 17341, 17342, 17349, 1735, 17350, 17351, 17352, 17359, 1736, 17360, 17361, 17362, 17369, 1737, 17370, 17371, 17372, 17379, 1738, 17380, 17381, 17382, 17389, 1739, 17390, 17391, 17392, 17399, 174, 1740, 1741, 1742, 1743, 1744, 1745, 1746, 1748, 1749, 175, 1750, 1759, 176, 1760, 1761, 1762, 1763, 1764, 1765, 1768, 1769, 179, 180, 1800, 1801, 1808, 1809, 181, 182, 1820, 1821, 1828, 183, 1830, 1832, 1833, 1834, 1835, 1838, 1839, 184, 1840, 1841, 1842, 1843, 1844, 1848, 1849, 185, 186, 1860, 1869, 187, 1871, 1872, 1873, 1874, 1875, 1876, 1877, 1878, 1879, 188, 1880, 1881, 1882, 1883, 1884, 1885, 1886, 1887, 1888, 1889, 189, 1890, 1891, 1892, 1893, 1894, 1898, 1899, 200, 2000, 20000,

(Continued)

TABLE E2. Continued

Baseline comorbidities	ICD codes from index and prior admissions
	20001, 20002, 20003, 20004, 20005, 20006, 20007, 20008, 2001, 20010, 20011, 20012, 20013, 20014, 20015, 20016, 20017, 20018, 2002, 20020, 20021, 20022, 20023, 20024, 20025, 20026, 20027, 20028, 2003, 20030, 20031, 20032, 20033, 20034, 20035, 20036, 20037, 20038, 2004, 20040, 20041, 20042, 20043, 20044, 20045, 20046, 20047, 20048, 2005, 20050, 20051, 20052, 20053, 20054, 20055, 20056, 20057, 20058, 2006, 20060, 20061, 20062, 20063, 20064, 20065, 20066, 20067, 20068, 2007, 20070, 20071, 20072, 20073, 20074, 20075, 20076, 20076, 20077, 20078, 2008, 20080, 20081, 20082, 20083, 20084, 20085, 20086, 20087, 20088, 201, 2010, 20100, 20101, 20102, 20103, 20104, 20105, 20106, 20107, 20108, 2011, 20110, 20111, 20112, 20113, 20114, 20115, 20116, 20117, 20118, 2012, 20120, 20121, 20122, 20123, 20124, 20125, 20126, 20127, 20128, 2014, 20140, 20141, 20142, 20143, 20144, 20145, 20146, 20147, 20148, 2015, 20150, 20151, 20152, 20153, 20154, 20155, 20156, 20157, 20158, 2016, 20160, 20161, 20162, 20163, 20164, 20165, 20166, 20167, 20168, 2017, 20170, 20171, 20172, 20173, 20174, 20175, 20176, 20177, 20178, 2019, 20190, 20191, 20192, 20193, 20194, 20195, 20196, 20197, 20198, 202, 2020, 20200, 20201, 20202, 20203, 20204, 20205, 20206, 20207, 20208, 2021, 20210, 20211, 20212, 20213, 20214, 20215, 20216, 20217, 20218, 2022, 20220, 20221, 20222, 20223, 20224, 20225, 20226, 20227, 20228, 2023, 20230, 20231, 20232, 20233, 20234, 20235, 20236, 20237, 20238, 2024, 20240, 20241, 20242, 20243, 20244, 20245, 20246, 20247, 20248, 2025, 20250, 20251, 20252, 20253, 20254, 20255, 20256, 20257, 20258, 2026, 20260, 20261, 20262, 20263, 20264, 20265, 20266, 20267, 20268, 2027, 20270, 20271, 20272, 20273, 20274, 20275, 20276, 20277, 20278, 2028, 20280, 20281, 20282, 20283, 20284, 20285, 20286, 20287,20288, 2029, 20290, 20291, 20292, 20293, 20294, 20295, 20296, 20297, 20298, 203, 2030, 20300, 20301, 20302, 2031, 20310, 20311, 20312, 2038, 20380, 20381, 20382, 204, 2040, 20400, 20401, 20402, 2041, 20410, 20411, 20412, 2042, 20420, 20421, 20422, 2048, 20480, 20481, 20482, 2049, 20490, 20491, 20492, 205, 2050, 20500, 20501, 20502, 2051, 20510, 20511, 20512, 2052, 20520, 20521, 20522, 2053, 20530, 20531, 20532, 2058, 20580, 20581, 20582, 2059, 20590, 20591, 20592, 206, 2060, 20600, 20601, 20602, 2061, 20610, 20611, 20612, 2062, 20620, 20621, 20622, 2068, 20680, 20681, 20682, 2069, 20690, 20691, 20692, 207, 2070, 20700, 20701, 20702, 2071, 20710, 20711, 20712, 2072, 20720, 20721, 20722, 2078, 20780, 20781, 20782, 208, 2080, 20800, 20801, 20802, 2081, 20810, 20811, 20812, 2082, 20820, 20821, 20822, 2088, 20880, 20881, 20882, 2089, 20890, 20891, 20892, 209, 2090, 20900, 20901, 20902, 20903, 2091, 20910, 20911, 20912, 20913, 20914, 20915, 20916, 20917, 2092, 20920, 20921, 20922, 20923, 20924, 20925, 20926, 20927, 20929, 2093, 20930, 20931, 20932, 20933, 20934, 20935, 20936, 2094, 20940, 20941, 20942, 20943, 2095, 20950, 20951, 20952, 20953, 20954, 20955, 20956, 20957, 2096, 20960, 20961, 20962, 20963, 20964, 20965, 20966, 20967, 20969, 2097, 20970, 20971, 20972, 20973, 20974, 20975, 20979, 190, 1900, 1901, 1902, 1903, 1904, 1905, 1906, 1907, 1908, 1909, 191, 1910, 1911, 1912, 1913, 1914, 1915, 1916, 1917, 1918, 1919, 192, 1920, 1921, 1922, 1923, 1928, 1929, 193, 194, 1940, 1941, 1943, 1944, 1945, 1946, 1948, 195, 1950, 1951, 1952, 1953, 1954, 1955, 1958, 196, 1960, 1961, 1962, 1963, 1965, 1966, 1968, 1969, 197, 1970, 1971, 1972, 1973, 1974, 1975, 1976, 1977, 1978, 198, 1980, 1981, 1982, 1983, 1984, 1985, 1986, 1987, 1988, 19881, 19882, 19889, 199, 1990, 1991 and 1992 V1001, V1002, V1021, V1003, V1004, V1005, V1006, V1007, V1000, V1009, V1011, V1012, V1020, V1022, V1082, V1083, V103, V1042, V1043, V1040, V1044, V1046, V1047, V1045, V1048, V1049, V1051, V1052, V1053, V1050, V1059, V1085, V1086, V1087, V1072, V1071, V1079, V1060, V1061, V1062, V1063, V1069, V1029, V1081, V1084, V1088, V1089, V109, V1090, V1091 and V711
	ICD-10-DX: C000, C001, C003, C004, C005, C006, C008,C002, C01, C020, C021, C022, C023, C024, C028, C029, C07, C080, C081, C089, C030, C031, C039, C040, C041, C048, C049, C060, C061, C050, C051, C052, C059, C062, C0689, C069, C099, C090, C091, C100, C101, C108, C102, C103, C104, C109, C110, C111, C112, C113, C118, C119, C130, C12, C131, C132, C138, C139,C140, C142, C148, C153, C154, C155, C158, C159, C160, C164, C163, C161, C162, C165, C166, C168, C169, C170, C171, C172, C173, C178, C179, C183, C184, C186, C187, C180, C181, C182, C185, C188, C189, C19, C20, C211, C210, C218, C220, C222, C227, C228, C221, C229, C240, C241, C248, C249, C250, C300, C301, C310, C311, C312, C313, C318, C319, C320, C321, C322, C323, C328, C329, C33, C3400, C3410, C342, C3430, C3480, C3490, C384, C37, C380, C381, C382, C388, C383, C390, C399, C251, C252, C253, C254, C257, C258, C259, C480, C481, C488, C482, C260, C261, C269, C410, C411, C412, C413, C4000, C4010, C414, C4020, C4030, C419, C490, C4910, C4920, C493, C494, C495, C496, C478, C498, C499, C430, D030, C4310, D0310, D0311, D0312, C4320, D0320, D0321, D0322, C4330, C4331, C4339, D0330, D0339, C434, D034, C4359, C0351, D0352, D0359, C4360, D0360, D0361, D0362, C4370, D0370, D0371, D0372, C438, D038, C439, D039, C4400, C4401, C4402, C4409, C44101, C44111,

(Continued)

TABLE E2. Continued

Baseline comorbidities	ICD codes from index and prior admissions
	C44121, C44191, C44201, C44211, C44221, C44291, C44300, C44301, C44309, C44310, C44311, C44319, C44320, C44321, C44329, C44390, C44391, C44399, C4440, C4441, C4442,C4449, C44500, C44501, C44509, C44510, C44511, C44519, C44520, C44521, C44529, C44590, C44591, C44599, C44601,C44611, C44621, C44691, C44701, C44711, C44721, C44791, C4480, C4481, C4482, C4489, C4490, C4491, C4492, C4499, C50019, C50119, C50219, C50319, C50419, C50519, C50619, C50819, C50919, C50029, C50929, C460, C461,C462, C464, C4650, C463, C467, C469, C55, C530, C531, C538, C539, C58, C541, C542, C543, C549, C540, C548, C569, C5700, C5710, C573, C5720, C574, C52, C510, C511, C512, C519, C577, C578, C579, C61, C6200, C6210, C6290, C600, C601, C602, C609, C6300, C6310, C632, C608, C637,C638, C639, C670, C671, C672, C673, C674, C675, C676, C677, C678, C679, C649, C659, C669, C680, C681, C688, C689, C8330, C8339, C8331, C8332, C8333, C8334, C8335, C8336, C8337, C8338, C8350, C8359, C8352, C8353, C8354, C8355, C8356, C8357, C8358, C8370, C8379, C8371, C8372, C8373, C8374, C8375, C8376, C8377, C8378, C8380, C8389, C8381, C8382, C8383, C8384, C8385, C8386, C8387, C8388, C8310, C8319, C8311, C8312, C8313, C8314, C8315, C8316, C8317, C8318, C8460, C8469, C8470, C8479, C8461, C8471, C8462, C8472, C8463, C8473, C8464, C8474, C8465, C8475, C8466, C8476, C8467, C8477, C8468, C8478, C8171, C8172, C8173, C8174, C8175, C8176, C8177, C8178, C8100, C8109, C8140, C8149, C8101, C8141, C8102, C8142, C8103, C8143, C8104, C8144, C8105, C8145, C8106, C8146, C8107, C8147, C8108, C8148, C8110, C8119, C8111, C8112, C8113, C8114, C8115, C8116, C8117, C8118, C8120, C8129, C8121, C8122, C8123, C8124, C8125, C8126, C8127, C8128, C8130, C8139, C8131, C8132, C8133, C8134, C8135, C8136, C8137, C8138, C8190, C8199, C8191, C8192, C8193, C8194, C8195, C8196, C8197, C8198, C8290, C8299, C8291, C8292, C8293, C8294, C8295, C8296, C8297, C8298, C8400, C8409, C8401, C8402, C8403, C8404, C8405, C8406, C8407, C8408, C8410, C8419, C8411, C8412, C8413, C8414, C8415, C8416, C8417, C8418, C96A, C9140, C9141, C960, C962, C8440, C8449, C8441, C8442, C8443, C8444, C8445, C8446, C8447, C8448, C8580, C8589, C8581, C8582, C8493, C8583, C8584, C8585, C8586, C8587, C8588, C964, C96Z, C969, C9001, C9002, C9010, C9011, C9012, C888, C9020, C9030, C9021, C9031, C9022, C9032, C9100, C9101, C9102, C9110, C9111, C9112, C91Z0, C91Z1, C91Z2, C9190, C9191, C9192, C9200, C9240, C9250, C9201, C9241, C9251, C9202, C9242, C9252, C9210, C9211, C91212, C9220, C9221, C9222, C9230, C9231, C9232, C92Z0, C92Z1, C92Z2, C9290, C9291, C9292, C9300, C9301, C9302, C9310, C9311, C9312, C9390, C9391, C9392, C93Z0, C93Z1, C93Z2, C9400, C9401, C9402, D45, C45, C9420, C9421, C9422, C9430, C9480, C9431, C9481, C9432, C9482, C9500, C9501, C9502, C9510, C9511, C9512, C9590, C9591, C9592, C7A019, C7A010, C7A011, C7A012, C7A029, C7A020, C7A021, C7A022, C7A023, C7A024, C7A025, C7A026, C7A00, C7A090, C7A091, C7A092, C7A093, C7A094, C7A095, C7A096, C7A098, C7A1, C4A0, C4A10, C4A20, C4A30, C4A39, C4A4, C4A60, C4A70, C4A59, C4A8, C4A9, D3A019, D3A010, D3A011, D3A012, D3A029, D3A020, D3A021, D3A022, D3A023, D3A024, D3A025, D3A026, D3A00, D3A090, D3A091, D3A092, D3A093, D3A094, D3A095, D3A096, D3A098, C7B00, C7B01, C7B020, C7B03, C7B04, C7B1, C7B09, C7B8, C6940, C6960, C6950, C6900, C6910, C6920, C6930, C6980, C6990, C710, C711, C712, C713, C714, C715, C716, C717, C718, C719, C7250, C700, C709, C720, C721, C701, C729, C73, C7490, C750, C751, C752, C753, C754, C755, C758, C760, C761, C762, C763, C7640, C7650, C768, C770, C771, C772, C773, C774, C775, C778, C779, C7800, C781, C782, C7839, C784, C785, C786, C787, C7889, C7900, C7911, C7919, C792, C7931, C7932, C7949, C7951, C7952, C7960, C7970, C7981, C7982, C7989, C800, C801, C802, Z85810, Z85818, Z85819, Z8521, Z8501, Z85028, Z85038, Z85048, Z8505, Z8500, Z85068, Z8507, Z8509, Z85118, V8512, Z8520, Z8522, Z85820, Z85828, Z853, Z8542, Z8543, Z8540, Z8544, Z8545, Z8546, Z8547, Z8548, Z8549, Z8551, Z85528, Z8553, Z8550, Z8554, Z8559, Z85841, Z85848, Z85850, Z8571, Z8579, Z85831, Z856, Z85238, Z8529, Z85830, Z85840, Z85858, Z8589, Z859, Z85020, Z85030, Z85040, Z85060, Z85110, Z85230, Z85520, Z85821, Z0389
Coagulation/platelets disorders	ICD-9-DX: 286, 2860, 2861, 2862, 2863, 2864, 2865, 28652, 28653, 28659, 2866, 2867, 2869, 287, 2870, 2871, 2872, 2873, 28730, 28731, 28732, 28733, 28739, 2874, 2875, 2878, 2879 ICD-10-DX: D67, D681, D682, D680, D68311, D68312, D68318, D65, D6832, D684, D688, D689, D690, D691, D692, D6949, D693, D6941, D6942, D693, D696, D698, D699

(Continued)

TABLE E2. Continued

Baseline comorbidities	ICD codes from index and prior admissions
Immobility	ICD-9-DX: 3005, 7993, 78072, 7812, 78199, 3440, 34400, 34401, 34402, 34403, 34404, 34409, 3441, 3443, 34430, 34431, 34432, 3448, 34481, 34489, 3449, 7812, 7197, V4984, V463 ICD-10-DX: M623, G82, G822, G8220, G8221, G825, G8250, G8251, G8252, G8253, G8254, G831, G8310, G8311, G8312, G8313, G8314, G8381, G8389, G839, R532, R54, R26, R260, R261, R262, R268, R269, R2689, R4181, R296, R298, Z736, Z740, Z7409, Z7401, Z993
Cognitive dysfunction	ICD-9-DX: 2940, 2941, 29410, 29411, 2942, 29420, 29421, 2930, 2931, 3310, 3312, 2900, 2901, 29010, 29011, 29012, 29013, 29021, 2902, 29020, 2903, 2904, 29040, 29041, 29042, 29043, 2908, 2909, 2948, 2949, 78093, 29282, 797, 3108, 31089, 3109, 3312, 3310, 33111, 33119, 33182, 3319, 78093 ICD-10-DX: F00, F000, F001, F002, F009, F01, F010, F011, F012, F013, F015, F018, F019, F0150, F0151, F02, F020, F022, F023, F024, F028, F0280, F0281, F03, F039, F0390, F0391, F04, R41, F039, F0390, F0391, F1817, F1827,F1897, F1917, F1927, F0789, F09, F068, F070, G30, G300, G301, G308, G309, G310, G3101, G3109, G311, G318, G319, G3183, R411, R412, R413, R4181
Supplemental oxygen requirement or home ventilator dependent	ICD-9-DX: V462, V461, V4611, V4612, V4613, V4614 ICD-10-DX: Z9981, Z991, Z9911, Z9912, J95850

ICD-9, International Classification of Diseases, 9th Revision; *ICD-10*, International Classification of Diseases, 10th Revision; *PCS*, procedure codes; *DX*, diagnosis codes.

TABLE E3. Definition of secondary end points and falsification end points

Outcomes	International Classification of Diseases codes
Stroke	ICD-9-DX: 99702, 431, 432, 4320, 4321, 4329, 43311, 43321, 43331, 43381, 43391, 43401, 43411, 43491, I97811, I97821, 430, 43301 ICD-10-DX: 19781, 19782, 197810, 197811, 197820, 197821, I60, I600, I6000, I6001, I6002, I601, I6010, I6011, I6012, I602, I603, I6030, I6031, I6032, I605, I6050, I6051, I6052, I606, I607, I608, I609, I61, I610, I611, I612, I613, I614, I615, I604, I616, G460, G461, G462, G463, G464, G465, G466, G467, I63, I630, I6300, I6301, I63011, I63012, I63013, I63019, I6302, I6303, I63031, I63032, I63033, I63039, I6309, I631, I6310, I6311, I63111, I63112, I63113, I63119, I6312, I6313, I63131, I63132, I63133, I63139, I6319, I632, I6320, I6321, I63211, I63212, I63213, I63219, I6322, I6323, I63231, I63232, I63233, I63239, I6329, I633, I6330, I6331, I63311, I63312, I63313, I63319, I6332, I63321, I63322, I63323, I63329, I6333, I63331, I63332, I63333, I63339, I6334, I63341, I63342, I63343, I63349, I6339, I634, I6340, I6341, I63411, I63412, I63413, I63419, I6342, I63421, I63422, I63423, I63429, I6343, I63431, I63432, I63433, I63439, I6344, I63441, I63442, I63443, I63449, I6349, I635, I6350, I6351, I63511, I63512, I63513, I63519, I6352, I63521, I63522, I63523, I63529, I6353, I63531, I63532, I63533, I63539, I6354, I63541, I63542, I63543, I63549, I6359, I636, I638, I6381, I6389, I639, I6781, I6782, I6789
Aortic valve reintervention (including any surgical or transcatheter/percutaneous procedures on the aortic valve)	ICD-9-PCS: 3521, 3522, 3511, 3501, 3505, 3506, 3596, 3595 ICD-10-PCS: 02RF07Z, 02RF08Z, 02RF0JZ, 02RF0KZ, 02RF37H, 02RF37Z, 02RF38H, 02RF38Z, 02RF3JH, 02RF3JZ, 02RF3KH, 02RF3KZ, 02RF47Z, 02RF48Z, 02RF4JZ, 02RF4KZ, 02QF0ZZ, 02QF3ZZ, 02QF4ZZ, 027F0ZZ, 027F3ZZ, 027F4ZZ, 027F04Z, 027F34Z, 027F44Z, 027F0DZ, 027F3DZ, 027F4DZ, 02NF0ZZ, 02NF3ZZ, 02NF4ZZ, 02UF07Z, 02UF0JZ, 02UF0KZ, 02UF37Z, 02UF3JZ, 02UF3KZ, 02UF47Z, 02UF4JZ, 02UF4KZ, 02BF0ZZ, 02BF3ZZ, 02BF4ZZ, 025F0ZZ, 025F3ZZ, 025F4ZZ, X2RF032, X2RF432, X2RF332, 02WF07Z, 02WF08Z, 02WF0JZ, 02WF0KZ, 02WF47Z, 02WF48Z, 02WF4JZ, 02WF4KZ
Heart failure–related readmission	ICD-9-DX: 39891, 428, 4280, 4281, 4282, 42820, 42821, 42822, 42823, 4283, 42830, 42831, 42832, 42833, 4284, 42840, 42841, 42842, 42843, 4289, 429, 4290, 4291, 4292, 4293, 4294, 4295, 4296, 4297, 42971, 42979, 4298, 42981, 42982, 42983, 42989, 4299 ICD-10-DX: I0981, I110, I130, I132, I501, I5020, I5021, I5022, I5023, I5030, I5031, I5032, I5033, I5040, I5041, I5042, I5043, I50810, I50811, I50812, I50813, I50814, I5082, I5083, I5084, I5089, I509, I5181, I97110, I97130, I97190
Permanent pacemaker insertion	ICD-9-DX: V4501, V5331 ICD-9-PCS: 3780, 3781, 3782, 3783, 0050, 0053 ICD-10-DX: Z950, Z45010, Z45018 ICD-10-PCS: 0JH60PZ, 0JH63PZ, 0JH80PZ, 0JH83PZ, 0JH604Z, 0JH634Z, 0JH804Z, 0JH834Z, 0JH605Z, 0JH635Z, 0JH805Z, 0JH835Z, 0JH606Z, 0JH636Z, 0JH806Z, 0JH836Z, 0JH607Z, 0JH637Z, 0JH807Z, 0JH837Z
Urinary tract infection	ICD-9-DX: 5990 ICD-10-DX: N390
Pneumonia	ICD-9-DX: 4800, 4801, 4802, 4803, 4808, 4809, 481, 4820, 4821, 4822, 48230, 48231, 48232, 48239, 48240, 48241, 48242, 48249, 48281, 48282, 48283, 48284, 48289, 4829, 4830, 4831, 4838, 4841, 4843, 4845, 4846, 4847, 4848, 485, 486, 4870 ICD-10-DX: B953, B960, B961, J09X1, J1000, J1001, J1008, J1100, J1108, J120, J121, J122, J123, J1281, J1289, J129, J13, J14, J150, J151, J1520, J15211, J15212, J1529, J153, J154, J155, J156, J157, J158, J159, J160, J168, J17, J180, J181, J182, J188, J189, J200, J8281, J8282, J84111, J84116, J84117, J842, J851, J852

ICD-9, International Classification of Diseases, 9th Revision; *ICD-10*, International Classification of Diseases, 10th Revision; *PCS*, procedure codes; *DX*, diagnosis codes.

TABLE E4. Association between procedure year and the choice of surgical versus transcatheter aortic valve replacement procedures

	Unmatched		Matched	
Year	**SAVR (n = 3771)**	**TAVR (n = 2679)**	**SAVR (n = 797)**	**TAVR (n = 797)**
2012	8.4 (317)	2.8 (74)	4.6 (37)	4.4 (35)
2013	11.9 (450)	3.7 (99)	5.7 (45)	5.4 (43)
2014	11.1 (417)	5.9 (159)	6.0 (48)	6.8 (54)
2015	14.1 (532)	9.0 (240)	10.9 (87)	10.4 (83)
2016	13.1 (495)	12.6 (337)	13.6 (108)	12.9 (103)
2017	14.8 (558)	16.5 (442)	16.8 (134)	16.9 (135)
2018	13.6 (513)	20.9 (559)	19.6 (156)	19.8 (158)
2019	13.0 (489)	28.7 (769)	22.8 (182)	23.3 (186)

SAVR, Surgical aortic valve replacement; *TAVR*, transcatheter aortic valve replacement.

TABLE E5. Full multivariable Cox model evaluating all-cause mortality difference between of surgical versus transcatheter aortic valve replacement procedures

Covariate	Hazard ratio	95% CI		*P* value
		Lower	Upper	
Treatment group				
SAVR	Ref	Ref	Ref	Ref
TAVR (first 6 mo)	1.24	0.96	1.60	.10
TAVR (after 6 mo)	2.50	1.90	3.28	<.001
Male sex	1.11	0.95	1.31	.18
Age	1.00	0.99	1.01	.82
Hypertension	1.09	0.75	1.57	.66
Hyperlipidemia	0.73	0.59	0.89	.002
Peripheral vascular disease	1.29	1.12	1.49	<.001
Prior stroke	1.31	1.03	1.65	.03
Coronary artery disease	1.14	0.90	1.44	.29
Congestive heart failure	1.20	0.96	1.50	.11
Cardiomyopathy	1.18	1.01	1.38	.04
Dialysis dependency	1.68	1.29	2.21	<.001
Chronic kidney disease	1.25	1.09	1.43	.002
Chronic obstructive pulmonary disease	1.28	1.10	1.50	.002
Liver disease	1.13	0.95	1.33	.16
Coagulopathy	1.33	1.13	1.57	<.001
Cancer	1.14	0.98	1.32	.08
Immobility	1.04	0.86	1.25	.69
Supplemental oxygen dependency	1.42	1.12	1.80	.004
Cognitive dysfunction	0.92	0.75	1.14	.45
Procedural year				
2012	Ref	Ref	Ref	Ref
2019	0.39	0.26	0.59	<.001
2018	0.53	0.38	0.73	<.001
2017	0.68	0.51	0.92	.01
2016	0.65	0.48	0.87	.004
2015	0.68	0.50	0.92	.01
2014	0.99	0.75	1.31	.95
2013	0.98	0.71	1.36	.90
Admission type				
Elective	Ref	Ref	Ref	Ref
Urgent	1.33	1.06	1.69	.02
Emergency	1.82	1.41	2.35	<.001
Charlson comorbidity index	1.05	1.03	1.08	<.001
Frailty score	5.93	2.83	12.42	<.001

SAVR, Surgical aortic valve replacement; *TAVR*, transcatheter aortic valve replacement.

TABLE E6. Rate of 30-day new permanent pacemaker implantation after surgical aortic valve replacement and transcatheter aortic valve replacement by era

Treatment group	2012-2014	2015-2017	2018-2019
SAVR	5.38 (7/130)	5.47 (18/329)	7.40 (25/338)
TAVR	12.30 (15/122)	9.66 (31/321)	8.14 (28/344)

SAVR, Surgical aortic valve replacement; *TAVR*, transcatheter aortic valve replacement.